# IE 622: Probability and Stochastic Processes

Lecture Notes (Transcribed)

IIT Bombay

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# 1 Probability Foundations

**Example 1.1** (Discrete probability space)**.** Tossing a coin, rolling a die.

**Discrete probability space:** $\Omega$ is a finite or countable sample space. For $A \subseteq \Omega$,

$$\mathbb{P}(A) = \sum_{\omega \in A} p_\omega.$$

Toss a coin with bias $p$, $n$ times. Then $\Omega = \{0,1\}^n$, $\omega = (\omega_1, \omega_2, \ldots, \omega_n)$,

$$p_\omega = p^{\omega_1+\omega_2+\cdots+\omega_n}(1-p)^{n-(\omega_1+\omega_2+\cdots+\omega_n)}.$$

For $A = \{\text{exactly } k \text{ heads}\}$:

$$\mathbb{P}(A) = p^k(1-p)^{n-k}\binom{n}{k}.$$

**Definition 1.1** (Probability space)**.** A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where:

- $\Omega$ is a non-empty set (sample space).
- $\mathcal{F}$ is a **sigma-algebra** of subsets of $\Omega$:
  - (i) $\Omega \in \mathcal{F}$.
  - (ii) If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.
  - (iii) If $A_1, A_2, \ldots, A_n \in \mathcal{F}$, then $\bigcup_n A_n \in \mathcal{F}$.
- $\mathbb{P} : \mathcal{F} \to [0,1]$ is a **probability measure** satisfying:

$$\mathbb{P}(\Omega) = 1, \qquad \mathbb{P}\left(\bigsqcup_{i=1}^{\infty} A_i\right) = \sum_n \mathbb{P}(A_n).$$

**Remark 1.1.** A discrete probability space can be subsumed into $(\Omega, \mathcal{F}, \mathbb{P})$ by taking $\mathcal{F} = 2^\Omega$ (the power set, i.e. the collection of all subsets of $\Omega$).

**Definition 1.2** (Generated sigma-algebra)**.** Let $\mathcal{S}$ be a collection of subsets of $\Omega$. The **sigma-algebra generated by** $\mathcal{S}$ is

$$\sigma(\mathcal{S}) = \bigcap_{\substack{F \supseteq \mathcal{S} \\ F \text{ sigma-algebra}}} F,$$

i.e. the smallest sigma-algebra containing $\mathcal{S}$.

**Definition 1.3** (Borel sigma-algebra)**.** For $\Omega = \mathbb{R}$:

$$\mathcal{B}(\mathbb{R}) = \sigma\big(\{(a,b) : a < b,\ a, b \in \mathbb{R}\}\big).$$

More generally, $\mathcal{B}(\Omega) = \sigma(\{\text{open sets of } \Omega\})$.

**Definition 1.4** (Random variable)**.** Given $(\Omega, \mathcal{F}, \mathbb{P})$, a function $X : \Omega \to \mathbb{R}$ is a **random variable** if for every $x \in \mathbb{R}$,

$$\{\omega : X(\omega) \le x\} = X^{-1}((-\infty, x]) \in \mathcal{F}.$$

**Example 1.2.** Let $\Omega = \{1, 2, \ldots, 6\}$, $\mathcal{F} = \{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, \Omega\}$, $X : \Omega \to \mathbb{R}$ with

$$X(\omega) = \begin{cases} 1 & \text{if } \omega \bmod 3 = 0 \\ 0 & \text{otherwise.} \end{cases}$$

Then $X^{-1}((-\infty, 0]) = \{1, 2, 4, 5\}$ and $\mathbb{P}(\{1, 2, 4, 5\}) = 0$.

**Definition 1.5** (Random process / Stochastic process)**.** A **random process** is a collection $X = (X_t : t \in T)$ of random variables.

- $T$ finite $\Rightarrow$ random vectors.
- $T$ countable $\Rightarrow \{0, 1, 2, 3, \ldots\}$; $(X_1, X_2, \ldots)$ maps $(\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$.

**Definition 1.6** (Finite-dimensional distributions (fdd))**.** For any $m \in \{1, 2, 3, \ldots\}$ and finite set $\{t_1, t_2, \ldots, t_m\} \subset T$, define

$$f_{t_1, t_2, \ldots, t_m}(x_1, x_2, \ldots, x_m) = \mathbb{P}(X_{t_1} \le x_1,\ X_{t_2} \le x_2,\ \ldots,\ X_{t_m} \le x_m).$$

The collection $\{f_{t_1, \ldots, t_m} : m \in \{1, 2, \ldots\},\ (t_1, \ldots, t_m)\}$ is called the **set of fdd**.

Example: $f_{(X,Y)}(x, y) = \mathbb{P}(X \le x,\ Y \le y)$.

**Definition 1.7** (Consistency condition)**.** If $T_m \subset T_n \subset T$, then

$$f_{T_m}(x_1, x_2, \ldots, x_m) = f_{T_n}(\infty, \ldots, \infty, x_1, \infty, \ldots, \infty),$$

where $\infty$ is placed at the coordinates in $T_n \setminus T_m$.

**Theorem 1.1** (Kolmogorov Extension Theorem)**.** *If the fdd are consistent, then they define a unique stochastic process.*

# 2 Discrete-Time Markov Chains

## 2.1 Definitions and Basic Properties

**Definition 2.1** (Markov Chain (DTMC))**.** A DTMC $X \sim \text{Markov}(\lambda, P)$ is a stochastic process $(X_n;\ n \ge 0)$ with:

(i) **State space** $\mathcal{I}$: a countable set.

(ii) **Initial distribution** $\lambda = (\lambda_i : i \in \mathcal{I})$ with $\sum_{i \in \mathcal{I}} \lambda_i = 1$, $0 \le \lambda_i \le \infty$, and $\mathbb{P}(X_0 = i_0) = \lambda_{i_0}$.

(iii) **Transition probability matrix** (stochastic matrix) $P = (p_{ij},\ i, j \in \mathcal{I})$ with $\sum_j p_{ij} = 1$.

The process satisfies: for all $n > 0$ and $i_1, i_2, \ldots, i_{n+1} \in \mathcal{I}$,

$$\mathbb{P}(X_{n+1} = i_{n+1} \mid X_0 = i_0, X_1 = i_1, \ldots, X_n = i_n) = \mathbb{P}(X_{n+1} = i_{n+1} \mid X_n = i_n) = p_{i_n i_{n+1}}.$$

**Example 2.1** (Coin toss with bias $p$)**.** Let $X_n = 0$ (Tails) or 1 (Heads). Then $X_0, X_1, \ldots, X_n, X_{n+1}$ are i.i.d. and the process is Markov.

Define $Y_0 = 0$ and for $n \ge 1$, $Y_n = \sum_{i=0}^{n-1} X_i$ (number of heads before the $n$-th toss). Check: $(Y_n)_{n \ge 0}$ is a DTMC.

**Theorem 2.1** (Characterisation via fdd)**.** *$(X_n)_{n\geq 0} \sim$ Markov$(\lambda, P)$ if and only if for all $N$ and $i_0, \ldots, i_N \in \mathcal{I}$,*

$$\mathbb{P}(X_0 = i_0, X_1 = i_1, \ldots, X_N = i_N) = \lambda_{i_0} p_{i_0 i_1} p_{i_1 i_2} \cdots p_{i_{N-1} i_N}.$$

*Proof.* The forward direction follows from the Markov property by induction:

$$\begin{aligned}\mathbb{P}(X_0 = i_0, \ldots, X_n = i_n) &= \mathbb{P}(X_0 = i_0)\,\mathbb{P}(X_1 = i_1 \mid X_0 = i_0) \cdots \mathbb{P}(X_n = i_n \mid X_{n-1} = i_{n-1}) \\ &= \lambda_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}.\end{aligned}$$

For the conditional probability:

$$\begin{aligned}\mathbb{P}(X_{n+1} = i_{n+1} \mid X_0 = i_0, \ldots, X_n = i_n) &= \frac{\mathbb{P}(X_0 = i_0, \ldots, X_n = i_n, X_{n+1} = i_{n+1})}{\mathbb{P}(X_0 = i_0, \ldots, X_n = i_n)} \\ &= \frac{\lambda_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n} \cdot p_{i_n i_{n+1}}}{\lambda_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}} = p_{i_n i_{n+1}}.\end{aligned}$$

□

**Remark 2.1.** Write $\mathbb{P}(\,\cdot \mid X_0 = i) = \mathbb{P}_i(\,\cdot)$. Then $i \to j$ means $\mathbb{P}_i(X_n = j$ for some $n \geq 0) > 0$.

Define the Kronecker delta: $\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$ and $\delta_i = (\delta_{ij} : j \in \mathcal{I})$.

**Theorem 2.2** (Markov property at time $m$)**.** *Let $(X_n)_{n\geq 0} \sim$ Markov$(\lambda, P)$. Then conditional on $X_m = i$, the process $(X_{m+n})_{n\geq 0} \sim$ Markov$(\delta_i, P)$, and it is independent of $X_0, \ldots, X_m$.*

*Proof.* We show: for all $A \in \sigma(X_0, X_1, \ldots, X_m)$,

$$\mathbb{P}\big(\{X_m = i_m, \ldots, X_{m+n} = i_{m+n}\} \cap A \mid X_m = i\big) = \delta_{i,i_m} p_{i_m i_{m+1}} \cdots p_{i_{m+n-1} i_{m+n}}\, \mathbb{P}(A \mid X_m = i).$$

**Step 1:** Take $A = \{X_0 = i_0, \ldots, X_m = i_m\}$. Using the fdd formula, the LHS gives:

$$\frac{\lambda_{i_0} p_{i_0 i_1} \cdots p_{i_{m-1} i_m} \cdot p_{i_m i_{m+1}} \cdots p_{i_{m+n-1} i_{m+n}}}{\mathbb{P}(X_m = i)} = \mathbb{P}(A \cap \{X_m = i\}) \cdot \delta_{i,i_m} p_{i_m i_{m+1}} \cdots p_{i_{m+n-1} i_{m+n}} / \mathbb{P}(X_m = i).$$

**Step 2:** Extend to $A = \bigsqcup_{k=1}^{\infty} A_k$ by countable additivity. For $A = \{X_0 \in I_0, X_1 \in I_1, \ldots, X_m \in I_m\}$ where $I_0 = \{i_{01}, i_{02}\}$, we have $A = \{X_0 = i_{01}, \ldots, X_m = i_m\} \cup \{X_0 = i_{02}, \ldots, X_m = i_m\}$. The result follows by summing:

$$= \delta_{i,i_m} p_{i_m i_{m+1}} \cdots p_{i_{m+n-1} i_{m+n}} \sum_{k=1}^{\infty} \frac{\mathbb{P}(A_k \cap \{X_m = i\})}{\mathbb{P}(X_m = i)} = \mathbb{P}\left(\bigsqcup_{k=1}^{\infty} A_k \cap \{X_m = i\}\right). \qquad \square$$

□

**Theorem 2.3** (Chapman–Kolmogorov Equations)**.** *For any $n, l \geq 0$ and states $i, j \in \mathcal{I}$:*

$$p_{ij}^{(n+l)} = \sum_{k \in \mathcal{I}} p_{ik}^{(n)}\, p_{kj}^{(l)}.$$

*In matrix form: $P^{n+l} = P^n \cdot P^l$.*

*Proof.* By the law of total probability and the Markov property:

$$\begin{aligned} p_{ij}^{(n+l)} &= \mathbb{P}_i(X_{n+l} = j) \\ &= \sum_{k \in \mathcal{I}} \mathbb{P}_i(X_n = k)\, \mathbb{P}_i(X_{n+l} = j \mid X_n = k) \\ &= \sum_{k \in \mathcal{I}} p_{ik}^{(n)}\, \mathbb{P}_k(X_l = j) \quad \text{(Markov property at time } n) \\ &= \sum_{k \in \mathcal{I}} p_{ik}^{(n)}\, p_{kj}^{(l)}. \end{aligned}$$

The second equality uses the total probability law; the third uses the fact that given $X_n = k$, the future process $(X_{n+m})_{m \geq 0}$ is Markov$(\delta_k, P)$, independent of the past. □

**Remark 2.2.** The Chapman–Kolmogorov (CK) equation is the fundamental identity for computing $n$-step transition probabilities. It says: to go from $i$ to $j$ in $n + l$ steps, we must pass through some intermediate state $k$ after $n$ steps. The CK equations imply $P^n = P \cdot P^{n-1}$ (induction), so $p_{ij}^{(n)}$ can be computed as the $(i, j)$ entry of the matrix power $P^n$.

## 2.2 Class Structure

The idea is to break down the Markov chain into smaller entities that can be analysed independently.

**Definition 2.2** (Accessibility / Leads to)**.** $i \to j$ (*$i$ leads to $j$*, or $j$ is *reachable* from $i$) if $p_{ij}^{(n)} > 0$ for some $n \geq 0$, i.e. $\mathbb{P}_i(X_n = j$ for some $n \geq 0) > 0$.

**Definition 2.3** (Communication)**.** $i \leftrightarrow j$ (*$i$ communicates with $j$*) if $i \to j$ and $j \to i$.

**Proposition 2.1.** *$\leftrightarrow$ is an equivalence relation. Its equivalence classes are called* communicating classes.

**Remark 2.3.** $(P^2)_{ij} = \sum_{k \in \mathcal{I}} p_{ik} p_{kj}$.

**Definition 2.4** (Closed communicating class)**.** A communicating class $C$ is *closed* if for all $i \in C$ and $j \notin C$, $p_{ij} = 0$. (Once entered, cannot leave.)

**Remark 2.4** (Diagram)**.** Example with 4 states: States $\{1, 2, 3, 4\}$. States 1 and 2 communicate ($1 \leftrightarrow 2$), states 3 and 4 communicate ($3 \leftrightarrow 4$), with transitions from $\{1, 2\}$ to $\{3, 4\}$ possible. Class structure depends only on positive entries of $P$.

**Definition 2.5** (Absorbing state)**.** A set of states $A$ is *absorbing* if $A$ is a closed communicating class. If $A = \{i\}$, then $i$ is called an absorbing state (i.e. $p_{ii} = 1$).

**Definition 2.6** (Irreducibility)**.** $P$ (or the Markov chain $(X_n)$) is said to be *irreducible* if $\mathcal{I}$ is a single closed communicating class, i.e. $i \leftrightarrow j$ for all $i, j \in \mathcal{I}$.

## 2.3 Stopping Times and Strong Markov Property

**Definition 2.7** (Stopping time)**.** A random time $T : \Omega \to \{0, 1, 2, \ldots\} \cup \{\infty\}$ is a *stopping time* for $(X_n)_{n\geq 0} \sim \text{Markov}(\lambda, P)$ if for every $m \geq 0$,

$$\{T = m\} \in \sigma(X_0, X_1, \ldots, X_m),$$

i.e. the event $\{T = m\}$ depends only on $X_0, X_1, \ldots, X_m$.

**Example 2.2.** Let $T = \min\{n \geq 0 : X_n = 20\}$ (first time to reach state 20). Then

$$\{T = m\} = \{X_0 \neq 20, X_1 \neq 20, \ldots, X_{m-1} \neq 20, X_m = 20\} \in \sigma(X_0, \ldots, X_m).$$

So $T$ is a stopping time.

**Theorem 2.4** (Strong Markov Property)**.** *Let $(X_n)_{n\geq 0} \sim$ Markov$(\lambda, P)$ and let $T$ be a stopping time. Then conditional on $T < \infty$ and $X_T = i$, the process $(X_{T+n})_{n\geq 0} \sim$ Markov$(\delta_i, P)$, and it is independent of $X_0, \ldots, X_T$.*

*Proof.* We must show that for any $n \geq 0$ and states $j_0, j_1, \ldots, j_n$,

$$\mathbb{P}(X_{T+1} = j_1, \ldots, X_{T+n} = j_n \mid T < \infty, X_T = i, X_0, \ldots, X_{T-1}) = P^n(i, j_n) \cdot \mathbf{1}[j_0 = i].$$

Condition on $\{T = m\}$ for each $m$. On $\{T = m\}$, the event $\{X_T = i\}$ is $\{X_m = i\}$, which is $\sigma(X_0, \ldots, X_m)$-measurable. By the ordinary Markov property applied at fixed time $m$, the future $(X_{m+1}, X_{m+2}, \ldots)$ is independent of $(X_0, \ldots, X_m)$ given $X_m = i$ and follows Markov$(\delta_i, P)$. Summing over all $m$ gives the result. □

## 2.4 Hitting Times and Gambler's Ruin

**Definition 2.8** (Hitting time / First passage time)**.** For a state $j \in \mathcal{I}$, define the *first passage time* to $j$ as

$$T_j = \min\{n \geq 1 : X_n = j\}.$$

(If $X_n \neq j$ for all $n \geq 1$, set $T_j = \infty$.)

Define:

$$\begin{aligned} h_j^i &= \mathbb{P}_i(T_j < \infty) \quad \text{(probability of ever reaching } j \text{ starting from } i\text{)}, \\ m_j^i &= \mathbb{E}_i[T_j] \quad \text{(expected hitting time of } j \text{ from } i\text{)}. \end{aligned}$$

**Remark 2.5.** By convention, $\mathbb{P}_i(\cdot)$ means probability given $X_0 = i$, i.e. with initial distribution $\lambda = \delta_i$.

### 2.4.1 First Step Analysis

The key technique is to condition on $X_1$ (the first step) and use the Markov property. For the hitting probability $h_j^i$:

$$\begin{aligned} h_j^i &= \mathbb{P}_i(T_j < \infty) \\ &= \sum_{k\in\mathcal{I}} p_{ik}\, \mathbb{P}_i(T_j < \infty \mid X_1 = k) \\ &= p_{ij} + \sum_{k\neq j} p_{ik}\, h_j^k. \end{aligned}$$

This gives the system of equations:

$$h_j^i = p_{ij} + \sum_{k \neq j} p_{ik}\, h_j^k, \quad i \neq j,$$

with boundary condition $h_j^j = 1$.

Similarly, for expected hitting times $m_j^i$:

$$m_j^i = 1 + \sum_{k \neq j} p_{ik}\, m_j^k, \quad i \neq j,$$

with $m_j^j = 0$.

**Example 2.3** (Gambler's Ruin)**.** A gambler starts with \$$a$ ($0 < a < N$). At each step, with probability $p$ she wins \$1 and with probability $q = 1 - p$ she loses \$1. The game stops when she reaches \$0 (ruin) or \$$N$ (success). Model: $X_n$ = fortune at time $n$, $\mathcal{I} = \{0, 1, \ldots, N\}$, states 0 and $N$ are absorbing.

Let $h_i = \mathbb{P}_i(\text{reach } N \text{ before } 0)$. By first step analysis:

$$h_i = p\, h_{i+1} + q\, h_{i-1}, \quad 1 \leq i \leq N - 1,$$

with $h_0 = 0$ and $h_N = 1$.

**Case 1:** $p = q = 1/2$**.** The equation becomes $h_i = \frac{1}{2}h_{i+1} + \frac{1}{2}h_{i-1}$, i.e. $h_{i+1} - h_i = h_i - h_{i-1}$. So $h_i$ is linear: $h_i = \frac{i}{N}$.

**Case 2:** $p \neq q$**.** The characteristic equation of $ph_{i+1} - h_i + qh_{i-1} = 0$ has roots $r = 1$ and $r = q/p$. General solution: $h_i = A + B(q/p)^i$. Applying boundary conditions:

$$h_i = \frac{1 - (q/p)^i}{1 - (q/p)^N}.$$

**Theorem 2.5** (Minimality of Hitting Probabilities)**.** *Let $A \subseteq \mathcal{I}$ be a target set and define $h_i^A = \mathbb{P}_i(T_A < \infty)$ where $T_A = \min\{n \geq 0 : X_n \in A\}$. Then $\mathbf{h}^A = (h_i^A)_{i \in \mathcal{I}}$ is the* minimal non-negative solution *to:*

$$x_i = \begin{cases} 1 & i \in A, \\ \displaystyle\sum_{j \in \mathcal{I}} p_{ij}\, x_j & i \notin A. \end{cases}$$

*That is: (a) $\mathbf{h}^A$ satisfies the system, and (b) if $\mathbf{x} \geq 0$ also satisfies the system, then $x_i \geq h_i^A$ for all $i$.*

*Proof.* **(a) $h^A$ satisfies the system.** For $i \in A$: $h_i^A = \mathbb{P}_i(T_A < \infty) = 1$ (already there). For $i \notin A$, condition on the first step:

$$h_i^A = \mathbb{P}_i(T_A < \infty) = \sum_{j \in \mathcal{I}} p_{ij}\, \mathbb{P}_j(T_A < \infty) = \sum_{j \in \mathcal{I}} p_{ij}\, h_j^A.$$

**(b) Minimality.** Let $\mathbf{x} \geq 0$ satisfy the system. We show $x_i \geq h_i^A$ for all $i$ by induction on $n$: we prove $x_i \geq \mathbb{P}_i(T_A \leq n)$ for all $n \geq 0$.

*Base case* $n = 0$: $\mathbb{P}_i(T_A \leq 0) = \mathbf{1}_{i \in A} \leq x_i$ (since $x_i = 1$ for $i \in A$).

*Inductive step*: Assume $x_j \geq \mathbb{P}_j(T_A \leq n)$ for all $j$. For $i \notin A$:

$$x_i = \sum_j p_{ij} x_j \geq \sum_j p_{ij}\, \mathbb{P}_j(T_A \leq n) = \mathbb{P}_i(T_A \leq n + 1).$$

Taking $n \to \infty$: $x_i \geq \mathbb{P}_i(T_A < \infty) = h_i^A$. □

**Remark 2.6.** The minimality condition distinguishes $h^A$ from other solutions. For example, on an irreducible recurrent chain, $x_i \equiv 1$ also satisfies the system (with $A = \{j\}$), and $h_i^A = 1$ is indeed the minimal solution. For transient chains, other solutions with $x_i > h_i^A$ may exist. The linear system alone does not pin down $h^A$; the minimality condition is essential.

## 2.5 Recurrence and Transience

**Definition 2.9** (Recurrence and transience)**.** State $i$ is *recurrent* if $\mathbb{P}_i(X_n = i \text{ for infinitely many } n) = 1$, equivalently $h_i^i = 1$ (i.e. $T_i < \infty$ a.s.).

State $i$ is *transient* if $\mathbb{P}_i(X_n = i \text{ for infinitely many } n) = 0$, equivalently $h_i^i < 1$.

**Theorem 2.6** (Recurrence criterion via return visits)**.** *Let $V_i = \sum_{n=0}^{\infty} \mathbf{1}[X_n = i]$ be the total number of visits to $i$. Then:*

*(i) $i$ is recurrent $\iff \mathbb{E}_i[V_i] = \infty \iff \sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty$.*

*(ii) $i$ is transient $\iff \mathbb{E}_i[V_i] < \infty \iff \sum_{n=0}^{\infty} p_{ii}^{(n)} < \infty$.*

*Proof.* Note that $V_i = \sum_{n\geq 0} \mathbf{1}[X_n = i]$, so $\mathbb{E}_i[V_i] = \sum_{n\geq 0} p_{ii}^{(n)}$.

Let $f = h_i^i = \mathbb{P}_i(T_i < \infty)$. On each return to $i$, the chain restarts by the Strong Markov Property and returns again with probability $f$. So the number of visits $V_i$ has a geometric distribution: $\mathbb{P}_i(V_i \geq k) = f^{k-1}$ for $k \geq 1$.

If $f = 1$ (recurrent): $\mathbb{P}_i(V_i \geq k) = 1$ for all $k$, so $V_i = \infty$ a.s.

If $f < 1$ (transient): $\mathbb{P}_i(V_i \geq k) = f^{k-1} \to 0$, so $V_i < \infty$ a.s. and $\mathbb{E}_i[V_i] = 1/(1-f) < \infty$. □

**Theorem 2.7** (Class property of recurrence)**.** *Recurrence and transience are class properties: if $i \leftrightarrow j$ then $i$ is recurrent iff $j$ is recurrent.*

*Proof.* Suppose $i$ is recurrent and $i \leftrightarrow j$. Then there exist $m, n$ such that $p_{ij}^{(m)} > 0$ and $p_{ji}^{(n)} > 0$. For any $k \geq 0$:

$$p_{jj}^{(m+k+n)} \geq p_{ji}^{(n)}\, p_{ii}^{(k)}\, p_{ij}^{(m)}.$$

Summing over $k$: $\sum_k p_{jj}^{(m+k+n)} \geq p_{ji}^{(n)}\, p_{ij}^{(m)} \sum_k p_{ii}^{(k)} = \infty$, since $i$ is recurrent. Hence $j$ is recurrent. □

**Corollary 2.1.** *In a finite irreducible Markov chain, all states are recurrent.*

**Definition 2.10** (Positive recurrence and null recurrence)**.** A recurrent state $i$ is *positive recurrent* if $m_i^i = \mathbb{E}_i[T_i] < \infty$, and *null recurrent* if $m_i^i = \infty$.

**Theorem 2.8** (Recurrent class is closed)**.** *Every recurrent communicating class is a closed class.*

*Proof.* By contrapositive. Let $C$ be a communicating class that is not closed. Then $\exists\, i \in C,\, j \notin C$ with $p_{ij} > 0$. Since $j \notin C$, we have $j \not\to i$, so $\mathbb{P}(T_i = \infty \mid X_0 = i) \geq p_{ij} > 0$. Hence $f_{ii} = \mathbb{P}_i(T_i < \infty) < 1$, so $i$ is transient, contradicting recurrence of $C$. □

**Theorem 2.9** (Finite closed classes are recurrent)**.** *If $C$ is a finite closed communicating class, then all states in $C$ are recurrent.*

*Proof.* Suppose for contradiction some $i \in C$ is transient, so $f_{jj} < 1$ for all $j \in C$ (by class property). Let $V_j = \mathbb{E}_i[V_j]$ where $V_j = \#$ visits to $j$. For any $i, j \in C$:

$$\begin{aligned}\mathbb{E}_i[V_j] &= \sum_{n=1}^{\infty} \mathbb{P}_i(V_j \geq n) \\ &\leq 1 + \frac{f_{jj}}{1 - f_{jj}} = \frac{1}{1 - f_{jj}} < \infty.\end{aligned}$$

By the Strong Markov Property and summing over $j \in C$:

$$\sum_{j \in C} \mathbb{E}_i[V_j] = \sum_{j \in C} \frac{1}{1 - f_{jj}} < \infty.$$

But since $C$ is closed, the chain stays in $C$ forever, so $\sum_{j \in C} V_j = \infty$ a.s. Contradiction. □

## 2.6 Stationary Distributions

**Definition 2.11** (Stationary measure)**.** For a Markov chain $(X_n)_{n \geq 0}$ on $(\mathcal{I}, P)$, a row vector $\underline{\pi} = (\pi_i,\, i \in \mathcal{I})$ with $\pi_i \geq 0$ is a *stationary measure* if

$$\pi_j = \sum_{i \in \mathcal{I}} \pi_i \, p_{ij} \quad \forall\, j \in \mathcal{I}, \qquad \text{i.e.} \quad \underline{\pi} = \underline{\pi} P.$$

**Definition 2.12** (Stationary / invariant distribution)**.** A stationary measure $\underline{\pi}$ is a *stationary (invariant) distribution* if additionally $\sum_{i \in \mathcal{I}} \pi_i = 1$. Equivalently, $\underline{\pi} P^n = \underline{\pi}$ for all $n$.

**Theorem 2.10** (Stationarity is preserved over time)**.** *If $\underline{\pi}$ is a stationary distribution w.r.t. $P$ and $\mathbb{P}(X_0 = i) = \pi_i$, then $\mathbb{P}(X_m = i) = \pi_i$ for all $m$.*

*Proof.* By induction. The base case $m = 0$ holds by assumption. Assume $\mathbb{P}(X_m = j) = \pi_j$ for all $j$. Then:

$$\mathbb{P}(X_{m+1} = i) = \sum_{j \in \mathcal{I}} \mathbb{P}(X_{m+1} = i \mid X_m = j)\, \mathbb{P}(X_m = j) = \sum_{j \in \mathcal{I}} p_{ji} \, \pi_j = \pi_i.$$

□

**Example 2.4** (Two-state chain)**.** Let $\mathcal{I} = \{1, 2\}$ with

$$P = \begin{pmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{pmatrix}, \quad \alpha, \beta \in (0, 1).$$

The stationary distribution satisfies $\underline{\pi} P = \underline{\pi}$:

$$\underline{\pi} = \left( \frac{\beta}{\alpha + \beta}, \frac{\alpha}{\alpha + \beta} \right).$$

**Theorem 2.11** (Limit of transition probabilities is stationary)**.** *Let $\mathcal{I}$ be finite. Suppose for some $i \in \mathcal{I}$,*

$$p_{ij}^{(n)} \to \pi_j \quad \text{as } n \to \infty, \quad \forall\, j \in \mathcal{I}.$$

*Then $\underline{\pi} = (\pi_j : j \in \mathcal{I})$ is a stationary distribution.*

*Proof.* Since $\sum_j p_{ij}^{(n)} = 1$ and $|\mathcal{I}| < \infty$, taking $n \to \infty$ gives $\sum_j \pi_j = 1$. For stationarity:

$$\sum_{j \in \mathcal{I}} \pi_j \, p_{jk} = \lim_{n \to \infty} \sum_{j \in \mathcal{I}} p_{ij}^{(n)} \, p_{jk} = \lim_{n \to \infty} p_{ik}^{(n+1)} = \pi_k.$$

□

### 2.6.1 Stationary Measure via Cycle Formula

**Definition 2.13** (Expected number of visits per cycle)**.** For a Markov chain starting at $k$, and any $i \in \mathcal{I}$, define

$$\rho_i^k = \mathbb{E}_k\left[\sum_{n=0}^{T_k-1} \mathbf{1}\{X_n = i\}\right],$$

i.e. the expected number of visits to state $i$ during one excursion (cycle) from $k$ back to $k$.

**Theorem 2.12** (Cycle formula — stationary measure)**.** *Let $P$ be irreducible and recurrent. Then:*

*(a)* $\rho_k^k = 1$.

*(b)* $\underline{\rho}^k = (\rho_i^k : i \in \mathcal{I})$ *satisfies* $\underline{\rho}^k P = \underline{\rho}^k$.

*(c)* $0 < \rho_i^k < \infty$ *for all* $i \in \mathcal{I}$.

*Proof.* (a) Obvious: starting from $k$, we count the visit at time 0, so $\rho_k^k = 1$.

(b) We compute:

$$\rho_i^k = \mathbb{E}_k\left[\sum_{n=1}^{T_k} \mathbf{1}\{X_n = i\}\right] = \sum_{n=1}^{\infty} \mathbb{P}_k(X_n = i,\, T_k \geq n).$$

Using the Markov property at time $n-1$:

$$\begin{aligned}\rho_i^k &= \sum_{n=1}^{\infty}\sum_{j\in\mathcal{I}} p_{ji}\,\mathbb{P}_k(X_{n-1} = j,\, T_k \geq n)\\ &= \sum_{j\in\mathcal{I}} p_{ji} \sum_{m=0}^{\infty} \mathbb{P}_k(X_m = j,\, T_k > m) = \sum_{j\in\mathcal{I}} \rho_j^k\, p_{ji}.\end{aligned}$$

Hence $\underline{\rho}^k P = \underline{\rho}^k$.

(c) Since $P$ is irreducible, $\exists\, n, m$ s.t. $p_{ik}^{(m)} > 0$ and $p_{ki}^{(n)} > 0$. From (b), $\underline{\rho}^k P^m = \underline{\rho}^k$, so $\rho_i^k \geq p_{ki}^{(n)} \cdot \rho_k^k = p_{ki}^{(n)} > 0$. Also $\sum_j \rho_j^k = \mathbb{E}_k[T_k] = m_k < \infty$ if positive recurrent. □

**Lemma 2.1** (Uniqueness of stationary measure)**.** *Let $P$ be irreducible and $\underline{\pi}$ an invariant measure with $\pi_k = 1$. Then $\underline{\pi} \geq \underline{\rho}^k$ (componentwise). If $P$ is recurrent then $\underline{\pi} = \underline{\rho}^k$.*

**Definition 2.14** (Mean recurrence time)**.** For $i \in \mathcal{I}$, the *mean recurrence time* is $m_i = \mathbb{E}_i[T_i]$.

**Definition 2.15** (Positive and null recurrence)**.** A state $j \in \mathcal{I}$ is:

- *positive recurrent* if $f_{jj} = 1$ and $m_j < \infty$,
- *null recurrent* if $f_{jj} = 1$ and $m_j = \infty$.

**Example 2.5** (Simple symmetric random walk on $\mathbb{Z}$)**.** The SSRW ($p = q = 1/2$) on $\mathbb{Z}$ is irreducible. It is known that all states are recurrent ($f_{jj} = 1$) but null recurrent ($m_j = \infty$). Higher-dimensional symmetric random walks ($d \geq 3$) are transient.

**Theorem 2.13** (Expected Number of Visits Formula)**.** *Let $(X_n)_{n\geq 0} \sim$ Markov$(\delta_i, P)$. Define $f_{ij} = \mathbb{P}_i(T_j < \infty)$ and $f_{jj} = \mathbb{P}_j(T_j < \infty)$. Let $N_j = \sum_{n=0}^{\infty} \mathbf{1}[X_n = j]$ be the total number of visits to $j$. Then:*

$$\mathbb{E}_i[N_j] = \frac{f_{ij}}{1 - f_{jj}}.$$

*In particular, for $i = j$: $\mathbb{E}_j[N_j] = \frac{1}{1-f_{jj}}$ if $f_{jj} < 1$ (transient), and $\mathbb{E}_j[N_j] = \infty$ if $f_{jj} = 1$ (recurrent).*

*Proof.* Starting from $i$, the chain first reaches $j$ with probability $f_{ij}$. If it does reach $j$, then by the Strong Markov Property, from $j$ it makes subsequent returns to $j$, each with probability $f_{jj}$, independently.

Let $G$ = total visits to $j$ after the first visit. Then $G \sim \text{Geom}(1 - f_{jj})$ (number of successes before first failure, with success probability $f_{jj}$), so $\mathbb{E}[G] = f_{jj}/(1 - f_{jj})$.

Therefore:

$$\begin{aligned}\mathbb{E}_i[N_j] &= \underbrace{f_{ij}}_{\mathbb{P}(\text{ever visit } j)} \cdot \underbrace{\left(1 + \frac{f_{jj}}{1 - f_{jj}}\right)}_{\mathbb{E}[\text{visits to } j | \text{first visit made}]} \\ &= f_{ij} \cdot \frac{1}{1 - f_{jj}}.\end{aligned}$$

More formally: the number of visits to $j$ is $N_j = \sum_{k=1}^{\infty} \mathbf{1}[V_k]$ where $V_k$ is the event of a $k$-th visit to $j$. By the Strong Markov Property, $\mathbb{P}_i(V_k) = f_{ij} \cdot f_{jj}^{k-1}$ for $k \geq 1$. Hence:

$$\mathbb{E}_i[N_j] = \sum_{k=1}^{\infty} \mathbb{P}_i(V_k) = \sum_{k=1}^{\infty} f_{ij}\, f_{jj}^{k-1} = \frac{f_{ij}}{1 - f_{jj}}.$$

□

**Remark 2.7.** This formula also yields the recurrence/transience criterion: $\mathbb{E}_i[N_j] = \infty \iff f_{jj} = 1$ (for $i$ from which $j$ is reachable, i.e., $f_{ij} > 0$). Combined with $\mathbb{E}_i[N_j] = \sum_n p_{ij}^{(n)}$, this gives the divergence test: $j$ is recurrent iff $\sum_n p_{jj}^{(n)} = \infty$.

**Example 2.6** (Null Recurrence Despite Negative Expected Drift)**.** Consider the chain on $\mathcal{I} = \{0, 1, 2, \ldots\}$ (birth-death) with $p_0 = 1$ (reflecting barrier at 0) and for $i \geq 1$:

$$p_{i,i+1} = p_i, \quad p_{i,i-1} = 1 - p_i, \quad \text{with } p_i = \frac{1}{2} + \frac{c}{i} \text{ for some } c > 0.$$

The mean drift at state $i$ is $\mathbb{E}[X_{n+1} - X_n \mid X_n = i] = 2p_i - 1 = 2c/i > 0$ (positive drift!), yet for suitable $c$, this chain can be null recurrent.

More relevant to PS6 Q2: consider $p_i = 1/2 - c/i^2$, giving *negative drift* $-2c/i^2$ yet the chain is still null recurrent. The key point is:

**Negative drift does not imply positive recurrence.** The distinction is:

- **Transient:** $f_{jj} < 1$, equivalently $\sum_n p_{jj}^{(n)} < \infty$. The chain drifts to $+\infty$ and never returns.
- **Null recurrent:** $f_{jj} = 1$ but $m_j = \mathbb{E}_j[T_j] = \infty$. The chain always returns, but expected return time is infinite. Stationary distribution does NOT exist.

- **Positive recurrent:** $f_{jj} = 1$ and $m_j < \infty$. Stationary distribution exists.

A sufficient condition for null recurrence (not positive recurrence): by Foster–Lyapunov, the chain is recurrent if $\mathbb{E}[V(X_{n+1}) - V(X_n) \mid X_n = i] \leq 0$ for $i$ large (with $V(i) \to \infty$), but not positive recurrent if no solution to $\mathbb{E}[V(X_{n+1})] - V(i) \leq -\epsilon$ exists.

The simple symmetric random walk ($p_i = 1/2$, zero drift) on $\mathbb{Z}$ is the canonical null recurrent example: recurrent by Polya's theorem, yet $m_0 = \infty$.

**Theorem 2.14** (Characterisation of positive recurrence)**.** *Let $P$ be irreducible. The following are equivalent:*

*(i) Every state is positive recurrent.*

*(ii) Some state is positive recurrent.*

*(iii) $P$ has an invariant distribution $\underline{\pi}$; moreover if* (iii) *holds, $\pi_i = \dfrac{1}{m_i}$.*

*Proof sketch.* (i)⇒(ii) is obvious.

(ii)⇒(iii): Let $k$ be positive recurrent. Then $\underline{\rho}^k$ is a stationary measure with $\sum_j \rho_j^k = \mathbb{E}_k[T_k] = m_k < \infty$. Normalizing gives $\underline{\pi} = \underline{\rho}^k / m_k$.

(iii)⇒(i): If $\underline{\pi}$ is an invariant distribution for irreducible $P$, then $P$ is recurrent (by Lemma above, $\underline{\pi} = \underline{\rho}^k / m_k$ implies $m_k < \infty$). By class property, all states are positive recurrent and $\pi_i = 1/m_i$. □

## 2.7 Periodicity and Convergence

**Definition 2.16** (Period of a state)**.** The *period* of state $j \in \mathcal{I}$ is

$$d(j) = \gcd\{n \geq 1 : p_{jj}^{(n)} > 0\}.$$

A state $j$ is *aperiodic* if $d(j) = 1$.

**Example 2.7.** Consider the two-state chain with $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then $P^2 = I$, $P^{2n} = I$, $P^{2n+1} = P$, so $d(1) = d(2) = 2$ (periodic with period 2).

A state $j$ is aperiodic iff $p_{jj}^{(n)} > 0$ for all sufficiently large $n$.

**Lemma 2.2** (Aperiodicity is a class property)**.** *Suppose $P$ is irreducible and one state is aperiodic. Then for all $j, k \in \mathcal{I}$, $p_{jk}^{(n)} > 0$ for all sufficiently large $n$. In particular, all states are aperiodic.*

**Remark 2.8.** Periodicity is a class property: if $i \leftrightarrow j$ then $d(i) = d(j)$.

*Proof sketch (aperiodicity lemma).* Since $P$ is irreducible, $\exists\, r, s$ s.t. $p_{ji}^{(r)} > 0$ and $p_{ik}^{(s)} > 0$. Then

$$p_{jj}^{(r+n+s)} \geq p_{ji}^{(r)}\, p_{ii}^{(n)}\, p_{ik}^{(s)} > 0$$

for all sufficiently large $n$ (since $i$ is aperiodic). Hence $p_{jk}^{(r+n+s)} > 0$ for all large enough $n$. □

**Theorem 2.15** (Markov Chain Convergence Theorem)**.** *Let $P$ be irreducible and aperiodic, and suppose $\underline{\pi}$ is an invariant distribution for $P$. Then for any initial distribution $\lambda$,*

$$\mathbb{P}(X_n = j) \to \pi_j \quad \text{as } n \to \infty, \quad \forall\, j \in \mathcal{I}.$$

*In particular, $p_{ij}^{(n)} \to \pi_j$ as $n \to \infty$, for all $i, j$.*

*Proof (Coupling argument).* Define $(Y_n)_{n\geq 0} \sim \text{Markov}(\underline{\pi}, P)$ independently of $(X_n)$. Let $W_n = (X_n, Y_n)$; this is a Markov chain on $\mathcal{I} \times \mathcal{I}$ with transition probabilities

$$\tilde{P}_{(i,k)(j,\ell)} = p_{ij}\, p_{k\ell}.$$

The initial distribution of $W_0$ is $\mu_{(i,k)} = \lambda_i\, \pi_k$. Since $P$ is irreducible and aperiodic, $\tilde{P}$ is irreducible and positive recurrent with stationary distribution $\tilde{\pi}_{(i,k)} = \pi_i\, \pi_k$.

Fix $b \in \mathcal{I}$ and let $T = \inf\{n \geq 1 : X_n = Y_n = b\}$ (the *coupling time*).

**Step 1:** $\mathbb{P}(T < \infty) = 1$, since $\tilde{P}$ is positive recurrent (irreducible, finite or positive recurrent).

**Step 2:** Define the "stitched" process

$$Z_n = \begin{cases} X_n & \text{if } n < T, \\ Y_n & \text{if } n \geq T. \end{cases}$$

By the Strong Markov Property applied to $W_n$ at time $T$, $(Z_n)_{n\geq 0} \sim \text{Markov}(\lambda, P)$ and $(Z_n)$ has the same distribution as $(X_n)$.

**Step 3:**

$$\begin{aligned} |\mathbb{P}(X_n = j) - \mathbb{P}(Y_n = j)| &= |\mathbb{P}(Z_n = j) - \mathbb{P}(Y_n = j)| \\ &= |\mathbb{P}(X_n = j,\, n < T) - \mathbb{P}(Y_n = j,\, n < T)| \\ &\leq \mathbb{P}(n < T) \to 0 \quad \text{as } n \to \infty. \end{aligned}$$

Since $\mathbb{P}(Y_n = j) = \pi_j$ for all $n$ (stationarity), $\mathbb{P}(X_n = j) \to \pi_j$. □

## 2.8 Foster–Lyapunov Criterion

**Theorem 2.16** (Foster–Lyapunov Criterion)**.** *Let $(X_n)$ be a Markov chain on $\mathcal{I}$ with transition matrix $P$.*

*(i)* ***Recurrence:*** *$P$ is recurrent iff $\exists$ a function $V : \mathcal{I} \to \mathbb{R}_+$ and a finite set $A \subset \mathcal{I}$ such that:*

  *(a) $V(j) \to \infty$ as $j \to \infty$ (i.e. $V$ is a Lyapunov function),*

  *(b) $\forall\, i \notin A$: $\mathbb{E}[V(X_{m+1}) - V(X_m) \mid X_m = i] \leq 0$ (negative drift outside $A$).*

*(ii)* ***Transience:*** *$P$ is transient iff $\exists\, V : \mathcal{I} \to \mathbb{R}_+$ and finite $A \subset \mathcal{I}$ such that:*

  *(a) $\exists\, j \in A$ s.t. $V(j) < \inf_{i \notin A} V(i)$,*

  *(b) $\forall\, i \notin A$: $\mathbb{E}[V(X_{m+1}) - V(X_m) \mid X_m = i] \leq 0$.*

*The* mean drift *of $V$ from state $i$ in one step is:*

$$\mathbb{E}[V(X_{m+1}) - V(X_m) \mid X_m = i] = \sum_{j \in \mathcal{I}} p_{ij}(V(j) - V(i)).$$

**Example 2.8** (Random walk on $\mathbb{N}$)**.** Consider a random walk on $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$ with $p_{i,i+1} = p$ and $p_{i,i-1} = q$ for $i \geq 1$, absorbing at 0. Take $V(n) = n$ and $A = \{0\}$. For $i \geq 1$:

$$\mathbb{E}[V(X_{m+1}) - V(X_m) \mid X_m = i] = p(i+1) + q(i-1) - i = p - q.$$

If $p \leq q$ (i.e. $p \leq 1/2$), drift $\leq 0$ for $i \notin A$. By (i), the chain is recurrent.

For transience, take $V(n) = \alpha^n$ with $0 < \alpha < 1$, $A = \{0\}$. Then:

$$\mathbb{E}[V(X_{m+1}) - V(X_m) \mid X_m = i] = p\alpha^{i+1} + q\alpha^{i-1} - \alpha^i = \alpha^{i-1}(p\alpha^2 - \alpha + q).$$

This is $\leq 0$ iff $p\alpha^2 - \alpha + q \leq 0$. If $p > q$, choosing $\alpha = q/p < 1$ gives $p(q/p)^2 - q/p + q = q^2/p - q/p + q = q(q-1)/p + q \cdot (1 - 1/p + 1)$... the quadratic $p\alpha^2 - \alpha + q$ has roots $\alpha = 1$ and $\alpha = q/p$, so for $\alpha \in (q/p, 1)$ the expression is negative, confirming transience when $p > q$.

## 2.9 Positive Recurrence via Foster–Lyapunov: Additional Cases

Continuing the Foster–Lyapunov criterion: part (iii) of the theorem states that $P$ is *positive recurrent* iff there exists $V : \mathcal{I} \to \mathbb{R}_+$, $\epsilon > 0$, and a finite set $A \subset \mathcal{I}$ such that:

(a) $\forall i \in A$: $\mathbb{E}[V(X_{m+1}) \mid X_m = i] < \infty$.

(b) $\forall i \notin A$: $\mathbb{E}[V(X_{m+1}) - V(X_m) \mid X_m = i] \leq -\epsilon$.

**Example 2.9** (Positive recurrence: $V(x) = x$)**.** Take $V(x) = x$, $A = \{0\}$, drift $= p - q \leq -\epsilon$ for $p < q$. So $p \leq q \Rightarrow p < 1/2$ implies positive recurrence.

**Example 2.10** (Foster–Lyapunov with $V(i) = i$, non-homogeneous drift)**.** Let $p_{0,1} = 1$, $p_{i,i+1} = \frac{1}{2} - \frac{1}{2(i+1)}$ and $p_{i,i-1} = \frac{1}{2} + \frac{1}{2(i+1)}$ for $i \geq 1$. Take $V(i) = i$, $A = \{0\}$. Then:

$$\Delta V(i) = \mathbb{E}[V(X_{m+1}) - V(X_m) \mid X_m = i] = -\frac{1}{i+1}.$$

This $\to 0$ as $i \to \infty$; the Foster–Lyapunov drift condition fails for a fixed $\epsilon > 0$.

Take instead $V(i) = i^2$:

$$\mathbb{E}[X_{m+1}^2 - X_m^2 \mid X_m = i] = -\frac{i}{i+1} \leq -\frac{1}{2} \quad \text{for all } i \geq 1.$$

Foster–Lyapunov applies with $V(i) = i^2$ and $\epsilon = 1/2$, so the chain is positive recurrent.

## 2.10 Time Reversibility

**Definition 2.17** (Time Reversal)**.** Let $(X_n)_{n \geq 0} \sim \text{Markov}(\pi, P)$ where $\pi$ is the stationary distribution. For any $N \geq 1$, the *time-reversed chain* is $Y_n = X_{N-n}$, $0 \leq n \leq N$. Then $(Y_n)_{0 \leq n \leq N}$ is also a DTMC (by the Markov property applied in reverse).

*Derivation of reversed transition probabilities.*

$$\begin{aligned}
\mathbb{P}(Y_n = j \mid Y_{n+1} = i) &= \mathbb{P}(X_{N-n} = j \mid X_{N-n-1} = i) \\
&= \mathbb{P}(X_{N-n-1} = i \mid X_{N-n} = j) \frac{\mathbb{P}(X_{N-n} = j)}{\mathbb{P}(X_{N-n-1} = i)} \\
&= p_{ji} \frac{\mathbb{P}(X_{N-n} = j)}{\mathbb{P}(X_{N-n-1} = i)}.
\end{aligned}$$

If the chain starts in stationarity ($\lambda = \pi$), then $\mathbb{P}(X_n = j) = \pi_j$ for all $n$, so:

$$\mathbb{P}(Y_n = j \mid Y_{n+1} = i) = \frac{p_{ji}\,\pi_j}{\pi_i}.$$

The reversed chain has transition matrix $\hat{P}$ with $\hat{p}_{ij} = p_{ji}\pi_j/\pi_i$. □

**Definition 2.18** (Time Reversibility and Detailed Balance)**.** A DTMC with tpm $P$ and stationary distribution $\pi$ is *time reversible* if the reversed chain has the same tpm as the original, i.e.,

$$\pi_i\, p_{ij} = \pi_j\, p_{ji} \quad \forall\, i, j \in \mathcal{I}.$$

These are called the **Detailed Balance Equations (DBE)**.

**Remark 2.9.** If the DBE hold, then summing over $j$:

$$\sum_{j\in\mathcal{I}} \pi_i p_{ij} = \sum_{j\in\mathcal{I}} \pi_j p_{ji} \implies \pi_i = \sum_{j} \pi_j p_{ji} \implies \pi P = \pi.$$

So any $\pi$ satisfying the DBE is automatically a stationary distribution. This gives a much simpler route to finding $\pi$ in reversible chains: solve $\pi_i p_{ij} = \pi_j p_{ji}$.

**Example 2.11** (Birth-death chain: advantages of reversibility)**.** Consider a birth-death chain on $\{0, 1, 2, 3\}$ with $p_{i,i+1} = p$ and $p_{i,i-1} = q$ for $0 < i < 3$, reflecting at boundaries. The DBE give:

$$\pi_0 p = \pi_1 q, \quad \pi_1 p = \pi_2 q, \quad \pi_2 p = \pi_3 q.$$

So $\pi_1 = \pi_0(p/q)$, $\pi_2 = \pi_0(p/q)^2$, $\pi_3 = \pi_0(p/q)^3$. Normalizing:

$$\pi_0\Big(1 + (p/q) + (p/q)^2 + (p/q)^3\Big) = 1.$$

**Simple random walk on any graph is time reversible.**

# 3 Borel–Cantelli Lemma and Ergodic Theorem

## 3.1 Borel–Cantelli Lemma

**Definition 3.1** (Limsup and liminf of events)**.** For a sequence of events $\{A_n\}$:

$$\limsup_n A_n = \bigcap_{N\geq 1} \bigcup_{n\geq N} A_n = \{\omega : \omega \in A_n \text{ infinitely often}\} = \{A_n \text{ i.o.}\},$$
$$\liminf_n A_n = \bigcup_{N\geq 1} \bigcap_{n\geq N} A_n = \{\omega : \omega \notin A_n \text{ for only finitely many } n\}.$$

**Theorem 3.1** (Borel–Cantelli Lemma)**.** *Let $\{A_n\}$ be a sequence of events.*

*(i) If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$.*

*(ii) If $\{A_n\}$ are independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$.*

*Proof of part (i).* Let $B_N = \bigcup_{n \geq N} A_n$. Then $B_1 \supseteq B_2 \supseteq \cdots$ and $\bigcap_{N=1}^{\infty} B_N = \limsup_n A_n$. By continuity of probability from above:

$$\mathbb{P}(A_n \text{ i.o.}) = \lim_{N\to\infty} \mathbb{P}(B_N) = \lim_{N\to\infty} \mathbb{P}\Big(\bigcup_{n\geq N} A_n\Big) \leq \lim_{N\to\infty} \sum_{n=N}^{\infty} \mathbb{P}(A_n) = 0,$$

since $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$ implies the tail sums $\to 0$. □

**Remark 3.1.** Note: part (ii) requires independence. For DTMC: if $\sum_{n=1}^{\infty} p_{ii}^{(n)} < \infty$ (transient), then $\mathbb{P}_i(X_n = i \text{ i.o.}) = 0$ by BCL(i). But when $\sum_n p_{ii}^{(n)} = \infty$, the events $\{X_n = i\}$ are *not* independent (since $\mathbb{P}_k(X_n = k, X_{n+1} = k) \neq \mathbb{P}_k(X_n = k)\mathbb{P}_k(X_{n+1} = k)$), so BCL(ii) does not directly apply.

## 3.2 Ergodic Theorem for DTMCs

**Definition 3.2** (Empirical visit count)**.** For a DTMC $(X_n)_{n\geq 0}$, define:

$$V_i(n) = \sum_{k=0}^{n-1} \mathbf{1}\{X_k = i\} \quad \text{(number of visits to state } i \text{ before time } n\text{)}.$$

**Theorem 3.2** (Ergodic Theorem for DTMCs)**.** *Let $P$ be irreducible and let $\lambda$ be any initial distribution. If $(X_n)_{n\geq 0} \sim$ Markov$(\lambda, P)$, then:*

*(a)* $\mathbb{P}\left[\dfrac{V_i(n)}{n} \to \dfrac{1}{m_i} \text{ as } n \to \infty\right] = 1$, *where $m_i = \mathbb{E}_i[T_i]$ (with $1/\infty = 0$).*

*(b) In the positive recurrent case, for any bounded $f : \mathcal{I} \to \mathbb{R}$:*

$$\mathbb{P}\left(\frac{1}{n}\sum_{k=0}^{n-1} f(X_k) \to \bar{f} \text{ as } n \to \infty\right) = 1,$$

*where $\bar{f} = \sum_{i\in\mathcal{I}} \pi_i f_i$ and $\pi$ is the unique stationary distribution.*

*Proof.* **Part (a) — Transient case:** If $P$ is transient, then $V_i := \sum_{k=0}^{\infty} \mathbf{1}\{X_k = i\} < \infty$ a.s. (since $\mathbb{E}_i[V_i] = 1/(1 - f_{ii}) < \infty$, hence $V_i < \infty$ a.s.). Thus $V_i(n) \leq V_i$ and $V_i(n)/n \leq V_i/n \to 0$, so $1/m_i = 0$ (since $m_i = \mathbb{E}_i[T_i] = \infty$ for transient $i$).

**Part (a) — Recurrent case:** Fix state $i$ and let $T = T_i$. Since $P$ is recurrent, $\mathbb{P}(T < \infty) = 1$. Let $S_i^{(k)}$ denote the length of the $k$-th excursion to state $i$; by the Strong Markov Property, $S_i^{(1)}, S_i^{(2)}, \ldots$ are i.i.d. with mean $m_i = \mathbb{E}_i[T_i]$.

The number of complete excursions by time $n$ satisfies $V_i(n) \to \infty$ as $n \to \infty$ (recurrence). Moreover, the sum of excursion lengths gives the squeeze:

$$\frac{S_i^{(1)} + \cdots + S_i^{(V_i(n)-1)}}{V_i(n)} \leq \frac{n}{V_i(n)} \leq \frac{S_i^{(1)} + \cdots + S_i^{(V_i(n))}}{V_i(n)}.$$

By the Strong Law of Large Numbers for i.i.d. sequences, both sides converge a.s. to $m_i$. Hence $n/V_i(n) \to m_i$ a.s., i.e., $V_i(n)/n \to 1/m_i = \pi_i$ a.s.

**Part (b) — Positive recurrent case:** Since $P$ is positive recurrent, $\pi_i = 1/m_i > 0$ and $\sum_i \pi_i = 1$. For any $J \subseteq \mathcal{I}$:

$$\begin{aligned}\left|\frac{1}{n}\sum_{k=0}^{n-1} f(X_k) - \bar{f}\right| &= \left|\sum_{i\in\mathcal{I}}\left(\frac{V_i(n)}{n} - \pi_i\right) f_i\right| \\ &\le \sum_{i\in J}\left|\frac{V_i(n)}{n} - \pi_i\right| + \sum_{i\notin J}\left(\frac{V_i(n)}{n} + \pi_i\right) \\ &\le 2\sum_{i\in J}\left|\frac{V_i(n)}{n} - \pi_i\right| + 2\sum_{i\notin J}\pi_i.\end{aligned}$$

Given $\epsilon > 0$, choose $J$ finite so that $\sum_{i\notin J}\pi_i < \epsilon/4$. Then, since $V_i(n)/n \to \pi_i$ a.s. for each $i$, there exists $N = N(\omega)$ such that for $n \ge N(\omega)$:

$$\sum_{i\in J}\left|\frac{V_i(n)}{n} - \pi_i\right| < \epsilon/4.$$

Thus for $n \ge N(\omega)$, the whole expression is $< \epsilon$. Since $\epsilon$ was arbitrary, the convergence holds a.s. □

# 4 Continuous-Time Markov Chains (CTMCs)

## 4.1 Introduction and Setup

**Definition 4.1** (Continuous-time stochastic process)**.** A collection of random variables $(X_t)_{t\ge 0}$ taking values in a discrete set $\mathcal{S}$ (the state space) is a *continuous-time random process*. We assume the sample paths $t \mapsto X_t(\omega)$ are right-continuous.

The *law* of $(X_t)$ is determined by its finite-dimensional distributions:

$$\mathbb{P}\big(\{X_{t_1} = i_1,\, X_{t_2} = i_2, \ldots, X_{t_n} = i_n\}\big)$$

for all $t_1 < t_2 < \cdots < t_n$ and states $i_1, \ldots, i_n$.

**Definition 4.2** (Continuous-Time Markov Chain)**.** $(X_t)_{t\ge 0}$ is a *Continuous-Time Markov Chain (CTMC)* if for all $t > 0$:

$$(X_{t+s},\, s \ge 0) \perp\!\!\!\perp (X_r,\, r < t) \mid X_t.$$

Equivalently, for $t_1 < t_2 < \cdots < t_n < t < s$:

$$\mathbb{P}(X_s = j \mid X_t = i,\, X_{t_n} = i_n, \ldots, X_{t_1} = i_1) = \mathbb{P}(X_s = j \mid X_t = i).$$

**Example 4.1** (DTMC embedded in continuous time)**.** Any DTMC $(X_n)$ with $p_{ii} > 0$ for all $i$ can be embedded into a CTMC by holding at each state for an $\mathrm{Exp}(\lambda)$ random time.

## 4.2 Exponential Distribution and Memoryless Property

**Definition 4.3** (Exponential random variable)**.** A random variable $T : \Omega \to [0, \infty)$ has the *exponential distribution* with rate $\lambda > 0$, written $T \sim \mathrm{Exp}(\lambda)$, if $f_T(t) = \lambda e^{-\lambda t}$ for $t \ge 0$. Equivalently, $\mathbb{P}(T > t) = e^{-\lambda t}$.

The key property of the exponential distribution is the *memoryless property*:

$$\mathbb{P}(T > t + s \mid T > t) = \mathbb{P}(T > s) \quad \forall\, t, s \geq 0. \tag{$*$}$$

**Lemma 4.1** (Characterization of Exponential)**.** *$T$ satisfies $(*)$ if and only if $T \sim \mathrm{Exp}(\lambda)$ for some $\lambda > 0$.*

*Proof.* ($\Rightarrow$) is clear. ($\Leftarrow$): Let $g(t) = \mathbb{P}(T > t)$. Then $(*)$ gives $g(s+t) = g(s) \cdot g(t)$ for all $s, t \geq 0$. Since $T > 0$, $g(1/n) > 0$ for large enough $n$, and $g(1) = \big(g(1/n)\big)^n > 0$. Let $g(1) = e^{-\lambda}$. Then $g(p/q) = \big(g(1/q)\big)^p = e^{-\lambda p/q}$ for all $p, q \in \mathbb{Z}$. Since $g$ is decreasing and right-continuous, $g(t) = e^{-\lambda t}$ for all real $t \geq 0$. □

**Example 4.2** (Minimum of exponentials)**.** If $S_1 \sim \mathrm{Exp}(\lambda_1), \ldots, S_n \sim \mathrm{Exp}(\lambda_n)$ are independent, then

$$\min\{S_1, S_2, \ldots, S_n\} \sim \mathrm{Exp}(\lambda_1 + \lambda_2 + \cdots + \lambda_n).$$

## 4.3 Poisson Process

**Definition 4.4** (Poisson Process)**.** A *Poisson process* of rate $\lambda > 0$, written $\mathrm{PP}(\lambda)$, is a right-continuous increasing integer-valued process $(X_t)_{t \geq 0}$ with $X_0 = 0$, constructed as follows. Let $\{S_n\}_{n \geq 1}$ be i.i.d. $\mathrm{Exp}(\lambda)$ inter-arrival times. Set $J_n = S_1 + \cdots + S_n$ (time of $n$-th jump). Then:

$$X_t = n \quad \text{when } J_n \leq t < J_{n+1}.$$

**Theorem 4.1** (Markov Property of Poisson Process)**.** *Let $(X_t)$ be a PP($\lambda$). For any $s > 0$, $(X_{t+s} - X_s)_{t \geq 0}$ is a PP($\lambda$) and is independent of the past $(X_r)_{r \leq s}$.*

*Proof.* It suffices to show this conditioned on $\{X_s = i\}$. On $\{X_s = i\}$:

(i) $X_r = \sum_{k=1}^{i} \mathbf{1}\{J_k \leq r\}$ for $r < s$.

(ii) The shifted holding times $\tilde{S}_1 = S_{i+1} - (s - J_i)$, $\tilde{S}_2 = S_{i+2}$, etc. are i.i.d. $\mathrm{Exp}(\lambda)$ by the memoryless property, and independent of $\{S_1, \ldots, S_i\}$.

Hence $(X_{t+s} - X_s)_{t \geq 0}$ has i.i.d. $\mathrm{Exp}(\lambda)$ inter-arrival times, making it a $\mathrm{PP}(\lambda)$ independent of $(X_r)_{r \leq s}$. □

**Theorem 4.2** (Strong Markov Property of Poisson Process)**.** *Let $T$ be a stopping time for $(X_t)$. Conditioned on $\{T < \infty\}$, the process $(X_{T+t} - X_T)_{t \geq 0}$ is a PP($\lambda$) independent of $(X_s,\ s \leq T)$.*

**Definition 4.5** (Stationary and Independent Increments)**.** A process $(X_t)$ has *stationary increments* if the law of $(X_{t+s} - X_s)$ depends only on $t$ (not on $s$). It has *independent increments* if increments over disjoint intervals are independent.

**Theorem 4.3** (Characterization of Poisson Process)**.** *Let $(X_t)$ be an increasing integer-valued process with $X_0 = 0$. The following are equivalent:*

*(i) The inter-arrival times $\{S_n\}$ are i.i.d.* $\mathrm{Exp}(\lambda)$.

*(ii) $(X_t)$ has independent increments and, uniformly in $t$:*

$$\mathbb{P}(X_{t+h} - X_t = 0) = 1 - \lambda h + o(h),$$
$$\mathbb{P}(X_{t+h} - X_t = 1) = \lambda h + o(h),$$

*as $h \to 0$.*

*(iii)* $(X_t)$ *has independent and stationary increments and* $X_t \stackrel{d}{=} Poi(\lambda t)$.

*Proof:* $(i) \Rightarrow (ii)$. By the Markov property, $(X_t)$ has independent increments. For the increment probabilities, note:

$$\mathbb{P}(X_{t+h} - X_t \geq 1) = \mathbb{P}(S_1 \leq h) = 1 - e^{-\lambda h} = \lambda h + o(h),$$
$$\mathbb{P}(X_{t+h} - X_t \geq 2) \leq \mathbb{P}(S_1 < h, S_2 < h) = (1 - e^{-\lambda h})^2 = o(h).$$

So $\mathbb{P}(X_{t+h} - X_t = 1) = \lambda h + o(h)$ and $\mathbb{P}(X_{t+h} - X_t = 0) = 1 - \lambda h + o(h)$.

**Proof:** $(ii) \Rightarrow (iii)$ Let $p_j(t) = \mathbb{P}(X_t = j)$ for $j \geq 0$. By independent increments:

$$\begin{aligned} p_j(t+h) &= \sum_{i=0}^{j} p_{j-i}(t)\, \mathbb{P}(X_{t+h} - X_t = i) \\ &= p_j(t)(1 - \lambda h + o(h)) + p_{j-1}(t)(\lambda h + o(h)) + o(h). \end{aligned}$$

This gives:

$$\frac{p_j(t+h) - p_j(t)}{h} = -\lambda p_j(t) + \lambda p_{j-1}(t) + \frac{o(h)}{h}.$$

Letting $h \to 0$: $p_j'(t) = -\lambda p_j(t) + \lambda p_{j-1}(t)$, with $p_0(0) = 1$, $p_j(0) = 0$ for $j \geq 1$.

For $j = 0$: $p_0'(t) = -\lambda p_0(t)$, so $p_0(t) = e^{-\lambda t}$. For $j = 1$: $p_1'(t) = -\lambda p_1(t) + \lambda e^{-\lambda t}$, so $p_1(t) = (\lambda t)e^{-\lambda t}$. By induction: $p_j(t) = \dfrac{(\lambda t)^j e^{-\lambda t}}{j!}$, i.e., $X_t \sim \mathrm{Poi}(\lambda t)$.

Since $(X_t)$ satisfies (ii) with increments depending only on interval length, it also has stationary increments.

**Proof:** $(iii) \Rightarrow (i)$ Property (iii) determines all finite-dimensional distributions, and we know that a process satisfying (i) also satisfies (iii). Since (iii) determines the law, (i) follows. □

**Theorem 4.4** (Superposition of Poisson Processes)**.** *If* $N_1(t) \sim PP(\lambda_1)$ *and* $N_2(t) \sim PP(\lambda_2)$ *are independent, then their superposition (merger)* $N(t) = N_1(t) + N_2(t) \sim PP(\lambda_1 + \lambda_2)$.

*More generally, the superposition of* $k$ *independent Poisson processes with rates* $\lambda_1, \ldots, \lambda_k$ *is* $PP(\lambda_1 + \cdots + \lambda_k)$.

*Proof.* $N(t) = N_1(t) + N_2(t)$ has independent increments (inherited from $N_1, N_2$ independent). For the increment distribution:

$$N(t) - N(s) = (N_1(t) - N_1(s)) + (N_2(t) - N_2(s)),$$

the sum of independent $\mathrm{Poi}(\lambda_1(t-s))$ and $\mathrm{Poi}(\lambda_2(t-s))$ random variables, which is $\mathrm{Poi}((\lambda_1 + \lambda_2)(t-s))$. By the characterization theorem, $N \sim \mathrm{PP}(\lambda_1 + \lambda_2)$. □

**Theorem 4.5** (Splitting (Thinning) of Poisson Processes)**.** *Let* $N(t) \sim PP(\lambda)$. *Independently mark each arrival as type 1 with probability* $p$ *and type 2 with probability* $1-p$. *Let* $N_1(t)$ *and* $N_2(t)$ *count type-1 and type-2 arrivals respectively. Then:*

*(i)* $N_1(t) \sim PP(\lambda p)$ *and* $N_2(t) \sim PP(\lambda(1-p))$.

*(ii)* $N_1$ *and* $N_2$ *are* ***independent*** *Poisson processes.*

*Proof.* We verify the joint distribution. For disjoint time intervals and any $m, n \geq 0$:

$$\begin{aligned}
\mathbb{P}(N_1(t) = m,\ N_2(t) = n) &= \mathbb{P}(N(t) = m + n) \cdot \binom{m+n}{m} p^m (1-p)^n \\
&= \frac{e^{-\lambda t}(\lambda t)^{m+n}}{(m+n)!} \cdot \frac{(m+n)!}{m!\,n!} p^m (1-p)^n \\
&= \frac{e^{-\lambda p t}(\lambda p t)^m}{m!} \cdot \frac{e^{-\lambda(1-p)t}(\lambda(1-p)t)^n}{n!}.
\end{aligned}$$

This factors as $\mathbb{P}(N_1(t) = m) \cdot \mathbb{P}(N_2(t) = n)$, confirming independence and the marginal Poisson distributions. The same argument applies to increments over any interval, establishing independence of the processes (not just at fixed time $t$). □

**Example 4.3** (PS9 Q2 type: Splitting into independent streams)**.** A router receives packets as a PP($\lambda$). Each packet is independently routed to server A with probability $p$ and server B with probability $1 - p$. By the splitting theorem, arrivals to server A form a PP($\lambda p$) and to server B form a PP($\lambda(1 - p)$), and these two streams are *independent.* This independence is surprising (one might expect them to be negatively correlated), and is a direct consequence of the Poisson distribution.

## 4.4 Generator Matrix and Transition Rate Matrix

**Definition 4.6** (Generator (Infinitesimal Generator) of a CTMC)**.** Let $(X_t)_{t\geq 0}$ be a CTMC on state space $S$. The **generator matrix** (or **Q-matrix**) $Q = (q_{ij})_{i,j\in S}$ is defined by:

$$\begin{aligned}
&q_{ij} \geq 0 \quad \text{for } i \neq j \quad \text{(transition rate from } i \text{ to } j\text{)}, \\
&q_{ii} = -\sum_{j\neq i} q_{ij} \quad \text{(diagonal entries are negative)}.
\end{aligned}$$

The **total departure rate** from state $i$ is $\lambda_i = -q_{ii} = \sum_{j\neq i} q_{ij}$.

**Remark 4.1.** The $Q$-matrix is the continuous-time analogue of the one-step transition matrix $P$ for DTMCs. Informally, for small $h > 0$:

$$\mathbb{P}(X_{t+h} = j \mid X_t = i) \approx \begin{cases} q_{ij}\, h & j \neq i, \\ 1 + q_{ii}\, h & j = i. \end{cases}$$

Thus $P(h) \approx I + Qh$ for small $h$, motivating the name "infinitesimal generator."

**Definition 4.7** (Transition Probability Matrix)**.** For a CTMC with generator $Q$, the **transition probability matrix** $P(t) = (p_{ij}(t))$ is defined by

$$p_{ij}(t) = \mathbb{P}(X_{t+s} = j \mid X_s = i), \quad t \geq 0,\ i, j \in S.$$

By time-homogeneity, $P(t)$ does not depend on $s$.

## 4.5 Kolmogorov Equations

**Theorem 4.6** (Kolmogorov Backward Equation)**.** *For a CTMC with generator $Q$ and transition matrices $P(t)$:*

$$\frac{d}{dt}P(t) = Q\,P(t), \quad P(0) = I.$$

**Theorem 4.7** (Kolmogorov Forward Equation)**.** *Under appropriate regularity conditions:*

$$\frac{d}{dt}P(t) = P(t)\,Q, \quad P(0) = I.$$

*Proof sketch (Backward Equation).* Fix $t > 0$ and let $h \to 0$. By the Chapman–Kolmogorov equation:

$$P(t+h) = P(h)\,P(t).$$

Then:

$$\frac{P(t+h) - P(t)}{h} = \frac{P(h) - I}{h}\,P(t).$$

Using $P(h) = I + Qh + o(h)$, we get $\dfrac{P(h) - I}{h} \to Q$ as $h \to 0$. Hence:

$$P'(t) = Q\,P(t).$$

□

**Theorem 4.8** (Matrix Exponential Solution)**.** *The unique solution to both Kolmogorov equations is:*

$$P(t) = e^{Qt} := \sum_{n=0}^{\infty} \frac{(Qt)^n}{n!}.$$

**Remark 4.2.** The matrix exponential $e^{Qt}$ has the following key properties:

- $e^{Q\cdot 0} = I$ (initial condition satisfied).
- All entries of $e^{Qt}$ are non-negative (since $Q$-matrix yields valid probabilities).
- Each row of $e^{Qt}$ sums to 1 (stochastic matrix).
- $\frac{d}{dt}e^{Qt} = Q\,e^{Qt} = e^{Qt}\,Q$ (both equations satisfied simultaneously).

## 4.6 Stationary Distribution of CTMCs

**Definition 4.8** (Stationary Distribution for CTMC)**.** A probability distribution $\pi = (\pi_i)_{i\in S}$ is a **stationary distribution** for the CTMC with generator $Q$ if:

$$\pi Q = \mathbf{0} \quad \text{and} \quad \sum_{i\in S} \pi_i = 1.$$

Equivalently, in component form: for each $j \in S$,

$$\sum_{i\in S} \pi_i\,q_{ij} = 0, \quad \text{i.e.,} \quad \pi_j\,\lambda_j = \sum_{i\neq j} \pi_i\,q_{ij}.$$

This is the **global balance equation**: rate of leaving state $j$ = rate of entering state $j$.

**Remark 4.3** (Relation to DTMC Stationary Distribution)**.** For DTMCs, stationarity was $\pi P = \pi$. For CTMCs, stationarity is $\pi Q = 0$. If $\pi$ is stationary for the CTMC, then for any $t \geq 0$:

$$\pi\, P(t) = \pi\, e^{Qt} = \pi \left( \sum_{n=0}^{\infty} \frac{(Qt)^n}{n!} \right) = \pi + \pi Q\, t + \cdots = \pi.$$

since $\pi Q = 0$ implies all higher-order terms vanish too.

## 4.7 Birth–Death Processes

**Definition 4.9** (Birth–Death Process)**.** A CTMC on $S = \{0, 1, 2, \ldots\}$ is a **birth–death process** if transitions only occur between adjacent states: for $i \geq 0$,

$$q_{i,i+1} = \lambda_i \geq 0 \quad \text{(birth rate)}, \quad q_{i,i-1} = \mu_i \geq 0 \quad \text{(death rate)}, \quad q_{i,i} = -(\lambda_i + \mu_i).$$

All other off-diagonal entries are zero ($q_0$ has no death: $\mu_0 = 0$).

**Theorem 4.9** (Stationary Distribution of Birth–Death Process)**.** *For an irreducible birth–death process, the stationary distribution (if it exists) satisfies the* ***detailed balance equations****:*

$$\pi_i\, \lambda_i = \pi_{i+1}\, \mu_{i+1}, \quad i = 0, 1, 2, \ldots$$

*The solution is:*

$$\pi_n = \pi_0 \prod_{k=0}^{n-1} \frac{\lambda_k}{\mu_{k+1}}, \quad n \geq 1,$$

*where $\pi_0$ is determined by normalization $\sum_{n=0}^{\infty} \pi_n = 1$.*

*Proof.* The global balance equation for state $j$ in a birth–death chain reduces to:

$$(\lambda_j + \mu_j)\pi_j = \lambda_{j-1}\pi_{j-1} + \mu_{j+1}\pi_{j+1}.$$

For birth–death processes, this telescopes: summing the equation over states $\{0, 1, \ldots, j\}$ yields $\pi_j \lambda_j = \pi_{j+1}\mu_{j+1}$ (detailed balance). Hence:

$$\pi_n = \frac{\lambda_{n-1}}{\mu_n}\pi_{n-1} = \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_1}\pi_0.$$

□

## 4.8 The M/M/1 Queue

**Example 4.4** (M/M/1 Queue)**.** The **M/M/1 queue** models a single-server queueing system with:

- Poisson arrivals at rate $\lambda$ (Markovian inter-arrival times $\sim \text{Exp}(\lambda)$),
- Exponential service times $\sim \text{Exp}(\mu)$ (one server),
- Unlimited waiting room (infinite buffer), FCFS discipline.

The state $X_t$ = number of customers in the system at time $t$ is a birth–death CTMC with:

$$\lambda_n = \lambda \quad (n \geq 0), \qquad \mu_n = \mu \quad (n \geq 1), \qquad \mu_0 = 0.$$

The **traffic intensity** is $\rho = \lambda/\mu$.

**Theorem 4.10** (Stationary Distribution of M/M/1)**.** *The M/M/1 queue has a stationary distribution if and only if $\rho = \lambda/\mu < 1$. In that case:*

$$\pi_n = (1-\rho)\,\rho^n, \quad n = 0, 1, 2, \ldots$$

*(geometric distribution with parameter $\rho$).*

*Proof.* Apply the detailed balance equations: $\pi_n\,\lambda = \pi_{n+1}\,\mu$, so $\pi_{n+1} = (\lambda/\mu)\pi_n = \rho\,\pi_n$. By induction: $\pi_n = \rho^n\,\pi_0$. Normalization: $\sum_{n=0}^{\infty} \rho^n \pi_0 = \pi_0/(1-\rho) = 1$ requires $\rho < 1$, giving $\pi_0 = 1-\rho$. Hence $\pi_n = (1-\rho)\rho^n$. □

**Remark 4.4** (Performance Measures for M/M/1)**.** Under stationarity ($\rho < 1$), Little's Law ($L = \lambda W$) and the geometric stationary distribution give:

$$\begin{aligned}
L = \mathbb{E}[X] &= \sum_{n=0}^{\infty} n\pi_n = \frac{\rho}{1-\rho} \quad \text{(mean number in system)},\\
W = L/\lambda &= \frac{1}{\mu-\lambda} \quad \text{(mean sojourn time)},\\
L_q &= \frac{\rho^2}{1-\rho} \quad \text{(mean queue length, excluding in service)},\\
W_q &= \frac{\rho}{\mu-\lambda} \quad \text{(mean waiting time in queue)}.
\end{aligned}$$

**Example 4.5** (Poisson Process as Pure Birth Chain)**.** The Poisson process $N_t \sim \mathrm{Poi}(\lambda t)$ is the simplest birth–death chain with $\lambda_n = \lambda$ for all $n$ and $\mu_n = 0$. Its $Q$-matrix is:

$$Q = \begin{pmatrix} -\lambda & \lambda & 0 & 0 & \cdots \\ 0 & -\lambda & \lambda & 0 & \cdots \\ 0 & 0 & -\lambda & \lambda & \cdots \\ \vdots & & & \ddots & \end{pmatrix}.$$

**Example 4.6** (Tandem Queue)**.** A **tandem queue** consists of two servers in series with a single Poisson arrival stream at rate $\lambda$. Server 1 has $\mathrm{Exp}(\mu_1)$ service times; departures from server 1 feed server 2 with $\mathrm{Exp}(\mu_2)$ service times. The joint queue length $(X_t^1, X_t^2)$ is a CTMC on $\mathbb{Z}_+^2$.

## 4.9 Explosion and Minimal Process

**Definition 4.10** (Explosion Time)**.** For a CTMC $(X_t)$ with holding times $S_n \sim \mathrm{Exp}(\lambda_{Y_n})$ in state $Y_n$, the **explosion time** is:

$$\zeta = \lim_{n\to\infty} J_n = \sum_{n=0}^{\infty} S_n,$$

where $J_n = S_0 + S_1 + \cdots + S_{n-1}$ is the time of the $n$-th jump. If $\zeta < \infty$ with positive probability, the chain **explodes**. After explosion, we set $X_t = \infty$ (cemetery state), giving the **minimal process**.

**Remark 4.5.** The "minimal" process is the canonical construction: it runs until explosion, then stays at $\infty$. For a CTMC $(X_t)$ that is strong Markov, the event $\{T \leq t\}$ depends only on $(X_s, s \leq t)$.

## 4.10 Class Structure of CTMCs

**Definition 4.11** (Accessibility and Communication in CTMCs)**.** State $j$ is **accessible** from $i$ (written $i \to j$) if $\mathbb{P}_i(X_t = j \text{ for some } t) > 0$. States $i$ and $j$ **communicate** ($i \leftrightarrow j$) if $i \to j$ and $j \to i$. Communicating classes, irreducibility, and closed sets are defined as in DTMCs.

**Theorem 4.11** (Class Structure Equivalences)**.** *For states $i, j$ in a CTMC, the following are equivalent:*

*(i)* $i \to j$ *in* $(X_t)$.

*(ii)* $i \to j$ *in the embedded DTMC* $\{Y_n\}$.

*(iii)* $q_{i,i_1} q_{i_1,i_2} \cdots q_{i_{n-1},j} > 0$ *for some states* $i_1, \ldots, i_{n-1} \in S$.

*(iv)* $p_{ij}(t) > 0$ *for all* $t > 0$.

*(v)* $p_{ij}(t) > 0$ *for some* $t > 0$.

*Proof.* $(iv) \Rightarrow (v) \Rightarrow (i)$ are clear. $(i) \Rightarrow (ii)$ is an exercise.

$(ii) \Rightarrow (iii)$: There exist states $i_1, \ldots, i_{n-1}$ such that $p_{i,i_1} p_{i_1,i_2} \cdots p_{i_{n-1},j} > 0$, which is equivalent to $\frac{q_{i,i_1}}{q_i} \cdots \frac{q_{i_{n-1},j}}{q_{i_{n-1}}} > 0$.

$(iii) \Rightarrow (iv)$: Suppose $q_{ij} > 0$. By the Chapman–Kolmogorov equation:

$$p_{ij}(t) \geq \mathbb{P}_i(J_1 < t,\ X_{J_1} = j,\ S_2 > t) = \mathbb{P}_i(J_1 < t,\ X_{J_1} = j)\, \mathbb{P}_i(S_2 > t) > 0.$$

For a general chain of length $n$, iterate using Chapman–Kolmogorov: $p_{ij}(t) \geq p_{i,i_1}(t/n) \cdots p_{i_{n-1},j}(t/n) > 0$ for all $t > 0$. □

**Remark 4.6.** Condition (iv) shows that CTMCs have **no notion of periodicity**: once $i \to j$, the transition probability $p_{ij}(t)$ is strictly positive for *all* $t > 0$, not just multiples of some period.

## 4.11 Recurrence and Transience of CTMCs

**Definition 4.12** (Recurrence and Transience for CTMCs)**.** Let $(X_t) = \text{Markov}(\lambda_i, Q)$ and $\{Y_n\}$ the embedded DTMC ($Y_n = X_{J_n}$). The **first return time** to state $i$ is:

$$T_i = \inf\{t \geq J_1 : X_t = i\}.$$

State $i$ is **recurrent** if $\mathbb{P}_i(\{t : X_t = i\} \text{ is unbounded}) = 1$, equivalently $\mathbb{P}_i(T_i < \infty) = 1$ (or $q_i = 0$). State $i$ is **transient** if $q_i > 0$ and $\mathbb{P}_i(T_i < \infty) < 1$.

**Theorem 4.12** (Recurrence via Embedded DTMC)**.** *(i) If $i$ is recurrent in $\{Y_n\}$, then $i$ is recurrent for $(X_t)$.*

*(ii) If $i$ is transient for $\{Y_n\}$, then $i$ is transient for $(X_t)$.*

*(iii) Every state is either recurrent or transient.*

*(iv) Recurrence and transience are class properties.*

*Proof.* (i) Since $i$ is recurrent in $\{Y_n\}$, $N = \sup\{n : Y_n = i\} = \infty$ w.p. 1, so $X_{J_n} = i$ for infinitely many $n$. Hence $\{t : X_t = i\} \supset \{J_n : Y_n = i\}$ is unbounded w.p. 1.

(ii) If $i$ is transient in $\{Y_n\}$, then $N = \sup\{n : Y_n = i\} < \infty$ w.p. 1, so $\{t : X_t = i\} \subset [0, J_{N+1}]$, which is bounded w.p. 1.

(iii)–(iv) follow from the DTMC results and parts (i)–(ii). □

## 4.12 Positive Recurrence and Stationary Distributions for CTMCs

**Definition 4.13** (Positive Recurrence for CTMCs)**.** A recurrent state $i$ is **positive recurrent** if $m_i = \mathbb{E}_i[T_i] < \infty$; otherwise it is **null recurrent**.

**Theorem 4.13** (Stationary Distribution via Embedded DTMC)**.** *The following are equivalent for an irreducible CTMC:*

*(i) $\pi$ is a stationary distribution for $Q$ (i.e., $\pi Q = 0$).*

*(ii) $\mu P = \mu$ where $\mu_i = \pi_i q_i$ (i.e., $\mu$ is stationary for the embedded DTMC).*

*Proof.* $(\mu(P - I))_i = \sum_j \mu_j p_{ji} - \mu_i = \sum_j \pi_j q_j \frac{q_{ji}}{q_j} - \pi_i q_i = \sum_j \pi_j q_{ji} - \pi_i q_i = (\pi Q)_i$. So $\mu P = \mu \Leftrightarrow \pi Q = 0$. □

**Theorem 4.14** (Positive Recurrence and Stationary Distribution for Irreducible CTMCs)**.** *Suppose $Q$ is irreducible. The following are equivalent:*

*(i) Every state is positive recurrent.*

*(ii) Some state is positive recurrent.*

*(iii) $Q$ is non-explosive and there exists a stationary distribution $\pi$ for $Q$.*

*Moreover, if (iii) holds, then* $\pi_i = \dfrac{1}{m_i q_i}$.

*Proof sketch.* Irreducibility implies $q_i > 0$ for all $i$. $(i) \Rightarrow (ii)$: trivial. $(ii) \Rightarrow (iii)$: If state $i$ is positive recurrent in $\{Y_n\}$, then $(X_t)$ doesn't explode. Define:

$$\mu_j^i = \mathbb{E}_i\left[\int_0^{T_i} \mathbf{1}_{\{X_s = j\}}\, ds\right] = \gamma_j^i \cdot \frac{1}{q_j},$$

where $\gamma_j^i = \mathbb{E}_i\left[\sum_{n=0}^{\infty} \mathbf{1}_{\{Y_n = j,\, n < N_i\}}\right]$ counts expected visits to $j$ in the embedded DTMC before returning to $i$, and $N_i = \inf\{n \geq 1 : Y_n = i\}$.

Since $S_{n+1} \sim \mathrm{Exp}(q_j)$ independently, $\mu_j^i = \gamma_j^i / q_j$.

From the DTMC result, $\gamma^i P = \gamma^i$, which gives $\mu^i Q = 0$. Moreover:

$$\sum_j \mu_j^i = \mathbb{E}_i\left[\int_0^{T_i} 1\, ds\right] = \mathbb{E}_i[T_i] = m_i < \infty.$$

So $\pi_j = \mu_j^i / m_i$ is a stationary distribution.

$(iii) \Rightarrow (i)$: Fix $i$. Let $v_j = \frac{\pi_j q_j}{\pi_i q_i}$; then $vP = v$. By the DTMC result, $v_j \geq \gamma_j^i$ componentwise, so $m_i = \sum_j \mu_j^i = \sum_j \gamma_j^i / q_j \leq \sum_j v_j / q_j = \frac{1}{\pi_i q_i} < \infty$. □

## 4.13 Convergence to Equilibrium for CTMCs

**Theorem 4.15** (Convergence to Equilibrium)**.** *Let $Q$ be irreducible and non-explosive with stationary distribution $\pi$. Then for all $i, j \in S$:*

$$p_{ij}(t) \to \pi_j \quad \text{as } t \to \infty.$$

*(No aperiodicity condition is needed, unlike in DTMCs.)*

*Proof.* Fix $h > 0$ and let $Z_n = X_{nh}$. Then $\{Z_n\}$ is a DTMC with transition matrix $P(h) = e^{Qh}$. Since $(X_t)$ is irreducible and non-explosive, $\{Z_n\}$ is irreducible and aperiodic (since $p_{ii}(h) > 0$ for all $h$), and $\pi$ is a stationary distribution for $P(h)$.

By the DTMC convergence theorem: $p_{ij}(nh) \to \pi_j$ as $n \to \infty$.

For general $t$, write $t = nh + r$ with $0 \le r < h$:

$$|p_{ij}(t) - \pi_j| \le |p_{ij}(nh) - \pi_j| + |p_{ij}(t) - p_{ij}(nh)|.$$

The second term satisfies $|p_{ij}(t+h) - p_{ij}(t)| \le 1 - e^{-q_i h}$ (an exercise using the Kolmogorov equations). Given $\varepsilon > 0$, choose $h$ so that $1 - e^{-q_i h} < \varepsilon$, then choose $N$ large so that $|p_{ij}(nh) - \pi_j| < \varepsilon$ for $n \ge N$. Then $|p_{ij}(t) - \pi_j| < 2\varepsilon$ for all $t \ge Nh$. □

## 4.14 Ergodic Theorem for CTMCs

**Theorem 4.16** (Ergodic Theorem for CTMCs)**.** *Let $(X_t) = \text{Markov}(\lambda, Q)$ where $Q$ is irreducible and non-explosive. Then for every bounded $f : S \to \mathbb{R}$:*

$$\frac{1}{t}\int_0^t f(X_s)\, ds \xrightarrow{a.s.} \sum_{j \in S} \pi(j)\, f(j) \quad \text{as } t \to \infty,$$

*where the right side is the ensemble (spatial) average. The result follows from the Strong Markov Property and the Strong Law of Large Numbers.*

**Remark 4.7.** In particular, with $f = \mathbf{1}_{\{j\}}$:

$$\frac{1}{t}\int_0^t \mathbf{1}_{\{X_s = j\}}\, ds \xrightarrow{a.s.} \pi_j = \frac{1}{m_j q_j},$$

which says the long-run fraction of time spent in state $j$ equals $\pi_j$.

## 4.15 Time Reversibility of CTMCs

**Definition 4.14** (Time-Reversed CTMC)**.** Let $(X_t) = \text{Markov}(\pi, Q)$ where $Q$ is irreducible, non-explosive, with stationary distribution $\pi$. Fix $T > 0$ and define the **time-reversed process** $\hat{X}_t = X_{T-t}$. The reversed process has generator $\hat{Q} = (\hat{q}_{ij})$ given by:

$$\hat{q}_{ji} = \frac{\pi_i}{\pi_j}\, q_{ij}, \quad i \ne j.$$

**Theorem 4.17** (Reversed CTMC)**.** *Let $(X_t) = \text{Markov}(\pi, Q)$ with $\pi Q = 0$. Then:*

*(i) $\hat{Q}$ is a valid rate matrix, irreducible and non-explosive.*

*(ii) $\pi \hat{Q} = 0$ (i.e., $\pi$ is stationary for $\hat{Q}$).*

*(iii)* $(\hat{X}_t) = Markov(\pi, \hat{Q})$.

*Proof sketch.* For (i): $\sum_{i\neq j} \hat{q}_{ji} = \sum_{i\neq j} \frac{\pi_i}{\pi_j} q_{ij}$. Since $\pi Q = 0$, $\sum_i \pi_i q_{ij} = 0$, giving $\pi_j q_{jj} = -\sum_{i\neq j} \pi_i q_{ij}$, so $\sum_{i\neq j} \hat{q}_{ji} = \frac{-\pi_j q_{jj}}{\pi_j} = -q_{jj} = q_j$. Hence $\hat{q}_{jj} = -q_j$ and $\hat{Q}$ is a valid rate matrix.

For (ii): $(\pi\hat{Q})_i = \sum_j \pi_j \hat{q}_{ji} = \sum_{j\neq i} \pi_j \frac{\pi_i}{\pi_j} q_{ji} + \pi_i \hat{q}_{ii} = \pi_i \sum_{j\neq i} q_{ji} - \pi_i q_i = \pi_i(\sum_j q_{ji} - q_i) = 0$ since $\sum_j q_{ji} = 0$.

For (iii): One verifies the finite-dimensional distributions of $\hat{X}$ match those of Markov$(\pi, \hat{Q})$ by direct calculation using the stationarity of $\pi$. □

**Definition 4.15** (Reversibility for CTMCs)**.** $(X_t)$ started in stationarity is **reversible** w.r.t. $\pi$ if $(\hat{X}_t) \stackrel{d}{=}$ Markov$(\pi, Q)$, i.e., $\hat{Q} = Q$. This is equivalent to the **detailed balance equations**:

$$\pi_i\, q_{ij} = \pi_j\, q_{ji}, \quad \forall\, i \neq j.$$

**Remark 4.8.** Detailed balance $\Rightarrow$ global balance ($\pi Q = 0$). Indeed:

$$(\pi Q)_i = \sum_j \pi_j q_{ji} = \sum_{j\neq i} \pi_j q_{ji} + \pi_i q_{ii} = \sum_{j\neq i} \pi_i q_{ij} - \pi_i q_i = 0.$$

The M/M/1 queue satisfies detailed balance: $\pi_n \lambda = \pi_{n+1}\mu$ gives $\pi_{n+1} = \rho\pi_n$, yielding the geometric stationary distribution and confirming reversibility.

**Example 4.7** (M/M/1 is Reversible)**.** The M/M/1 queue with $\rho < 1$ satisfies detailed balance: $\pi_n\lambda = \pi_{n+1}\mu$ for all $n \geq 0$. Hence it is time-reversible under $\pi_n = (1-\rho)\rho^n$. The reversed process is again an M/M/1 queue with the same rates.

## 4.16 Burke's Theorem

**Theorem 4.18** (Burke's Theorem)**.** *Consider an M/M/1 queue with arrival rate $\lambda$ and service rate $\mu$, with $\rho = \lambda/\mu < 1$, started in stationarity (i.e., $X_0 \sim \pi$). Then:*

*(i) The departure process $D(t)$ is a Poisson process with rate $\lambda$.*

*(ii) For each $t \geq 0$, the number of customers in the system $X_t$ is independent of the departure process $\{D(s) : 0 \leq s \leq t\}$.*

*Proof Sketch.* Since the M/M/1 queue is time-reversible under $\pi$, the reversed process $\hat{X}_t = X_{T-t}$ is also an M/M/1 Markov chain with the same transition rates. In the reversed process, arrivals and departures interchange roles: departures in the forward process correspond to arrivals in the reversed process. Since arrivals in the reversed process form a PP($\lambda$) (by the M/M/1 input), departures in the forward process also form a PP($\lambda$). Independence of $X_t$ and past departures follows from the Markov property applied to the reversed chain. □

**Remark 4.9.** Burke's theorem is a consequence of time-reversibility. It is the key tool for analyzing networks of queues (Jackson networks), since it implies that the output of each M/M/1 node in stationarity is Poisson and independent of the node's state.

## 4.17 Tandem Queues

**Example 4.8** (Tandem Queue)**.** Consider two M/M/1 queues in series: customers arrive at queue 1 at rate $\lambda$ (Poisson), are served at rate $\mu_1$, depart and immediately enter queue 2 where they are served at rate $\mu_2$. Let $\rho_k = \lambda/\mu_k < 1$ for $k = 1, 2$.

By Burke's theorem, the departure process from queue 1 is PP($\lambda$), independent of $X_1(t)$ (the number in queue 1). Hence the input to queue 2 is also PP($\lambda$), independent of the state of queue 1. The joint stationary distribution is:

$$\pi(n_1, n_2) = \pi_1(n_1)\,\pi_2(n_2) = (1-\rho_1)\rho_1^{n_1} \cdot (1-\rho_2)\rho_2^{n_2}, \quad n_1, n_2 \geq 0.$$

This product-form result says the two queues in equilibrium behave as if they were independent M/M/1 queues.

**Remark 4.10.** The independence in the tandem queue is non-trivial: even though departures from queue 1 feed queue 2, the two queue lengths are independent in stationarity. This is a direct consequence of Burke's theorem and fails without time-reversibility.

## 4.18 Jackson Networks

**Definition 4.16** (Open Jackson Network)**.** An **open Jackson network** consists of $K$ nodes (queues) where:

- External customers arrive to node $i$ according to PP($d_i$), $d_i \geq 0$.
- Node $i$ has a single server with exponential service time at rate $\mu_i$.
- Upon service completion at node $i$, a customer routes to node $j$ with probability $p_{ij}$, or leaves the network with probability $1 - \sum_j p_{ij}$.

**Definition 4.17** (Traffic Equations)**.** The **effective arrival rate** $\beta_i$ at node $i$ satisfies the **traffic equations**:

$$\beta_i = d_i + \sum_{j=1}^{K} \beta_j\, p_{ji}, \quad i = 1, \ldots, K.$$

Let $\rho_i = \beta_i/\mu_i$. The network is stable if $\rho_i < 1$ for all $i$.

**Theorem 4.19** (Existence and Uniqueness of Traffic Equation Solution)**.** *Let $P = (p_{ij})$ be the routing matrix with spectral radius $r(P) < 1$ (i.e., the network is open: every customer eventually leaves). Then the traffic equations $\beta_i = d_i + \sum_j \beta_j p_{ji}$, $i = 1, \ldots, K$, have a unique non-negative solution given by:*

$$\boldsymbol{\beta} = \mathbf{d}(I-P)^{-1} = \mathbf{d} + \mathbf{d}P + \mathbf{d}P^2 + \cdots$$

*Proof.* In matrix form the equations are $\boldsymbol{\beta} = \mathbf{d} + \boldsymbol{\beta}P$, i.e., $\boldsymbol{\beta}(I-P) = \mathbf{d}$. The matrix $(I-P)$ is invertible if and only if $r(P) < 1$ (spectral radius strictly less than 1). Since the network is open, every customer eventually leaves, so $P^n \to 0$ entry-wise as $n \to \infty$, and the Neumann series converges:

$$(I-P)^{-1} = \sum_{n=0}^{\infty} P^n.$$

Therefore $\boldsymbol{\beta} = \mathbf{d}(I - P)^{-1}$ exists and is unique. The component $\beta_i = \sum_{n\geq 0}(\mathbf{d}P^n)_i$ counts the total rate of flow into node $i$: $(\mathbf{d}P^n)_i$ is the rate of customers who originated externally and have been routed through exactly $n$ intermediate nodes before reaching $i$. Non-negativity follows from $\mathbf{d} \geq 0$ and $P \geq 0$. □

**Remark 4.11.** The stability condition $\rho_i = \beta_i/\mu_i < 1$ for all $i$ ensures each node's queue is stable. When $\mathbf{d} = 0$ (no external arrivals), the only solution is $\boldsymbol{\beta} = 0$, confirming that the open network requires external input.

**Theorem 4.20** (Jackson's Theorem)**.** *For an open Jackson network in steady state with $\rho_i < 1$ for all $i$, the stationary distribution has the* ***product form****:*

$$\pi(n_1, n_2, \ldots, n_K) = \prod_{i=1}^{K} \pi_i(n_i),$$

*where $\pi_i(n_i) = (1 - \rho_i)\,\rho_i^{n_i}$ is the marginal distribution of node $i$, identical to that of an M/M/1 queue with load $\rho_i$.*

*Proof Sketch.* The joint state space is $\mathbb{N}^K$. The joint process $(X_1(t), \ldots, X_K(t))$ is a CTMC. One verifies that $\pi(\mathbf{n}) = \prod_i \pi_i(n_i)$ satisfies the global balance equations for the network CTMC. This is done by checking that the product-form distribution satisfies the *partial balance* (or local balance) equations at each node. The proof uses the fact that Poisson arrivals and exponential services give a tractable structure. □

**Remark 4.12.** Jackson's theorem is remarkable: even though queues share customers (via routing), in stationarity they are independent M/M/1 queues with effective loads $\rho_i$. This product-form property extends to networks of general $M/M/c$ nodes.

**Example 4.9** (Jackson Network Computation)**.** A network with $K = 2$ nodes: external arrivals $d_1 = 3$, $d_2 = 1$; routing $p_{12} = 0.5$, $p_{21} = 0.2$, all other $p_{ij} = 0$; service rates $\mu_1 = 6$, $\mu_2 = 4$.

Traffic equations:

$$\beta_1 = 3 + 0.2\,\beta_2, \quad \beta_2 = 1 + 0.5\,\beta_1.$$

Solving: $\beta_2 = 1 + 0.5(3 + 0.2\beta_2) = 2.5 + 0.1\beta_2$, so $\beta_2 = 2.5/0.9 \approx 2.78$, and $\beta_1 = 3 + 0.2 \times 2.78 \approx 3.56$.

Loads: $\rho_1 = 3.56/6 \approx 0.59$, $\rho_2 = 2.78/4 \approx 0.69 < 1$. The network is stable.

# 5 Martingales

## 5.1 Conditional Expectation

**Definition 5.1** (Conditional Expectation)**.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{G} \subseteq \mathcal{F}$ a sub-$\sigma$-algebra. For an integrable random variable $X$ (i.e., $E[|X|] < \infty$), the *conditional expectation* $E[X \mid \mathcal{G}]$ is the unique (a.s.) $\mathcal{G}$-measurable random variable $Z$ satisfying

$$\int_A Z\,dP = \int_A X\,dP \quad \text{for all } A \in \mathcal{G}.$$

Existence and uniqueness follow from the Radon–Nikodým theorem.

**Theorem 5.1** (Properties of Conditional Expectation)**.** *Let $X, Y$ be integrable random variables and $\mathcal{G} \subseteq \mathcal{H} \subseteq \mathcal{F}$.*

*(i)* ***Linearity:*** $E[\alpha X + \beta Y \mid \mathcal{G}] = \alpha E[X \mid \mathcal{G}] + \beta E[Y \mid \mathcal{G}]$ *a.s.*

*(ii)* ***Tower Property:*** $E\big[E[X \mid \mathcal{H}] \mid \mathcal{G}\big] = E[X \mid \mathcal{G}]$ *a.s.*

*(iii)* ***Measurability:*** *If $X$ is $\mathcal{G}$-measurable, then $E[X \mid \mathcal{G}] = X$ a.s.*

*(iv)* ***Independence:*** *If $X$ is independent of $\mathcal{G}$, then $E[X \mid \mathcal{G}] = E[X]$ a.s.*

*(v)* ***Jensen's Inequality:*** *If $\phi$ is convex and $E[|\phi(X)|] < \infty$, then $\phi(E[X \mid \mathcal{G}]) \leq E[\phi(X) \mid \mathcal{G}]$ a.s.*

*(vi)* ***Monotonicity:*** *If $X \leq Y$ a.s., then $E[X \mid \mathcal{G}] \leq E[Y \mid \mathcal{G}]$ a.s.*

**Remark 5.1.** The tower property is also called the *law of total expectation.* When $\mathcal{G} = \{\emptyset, \Omega\}$ (trivial $\sigma$-algebra), $E[X \mid \mathcal{G}] = E[X]$. The tower property says: if you have less information ($\mathcal{G} \subseteq \mathcal{H}$), averaging again over less information gives the less-informed expectation.

**Example 5.1** (Discrete Conditional Expectation)**.** Let $X$ and $Y$ be discrete random variables. Then

$$E[X \mid Y = y] = \sum_x x\, P(X = x \mid Y = y).$$

In this case $E[X \mid Y]$ is the random variable $g(Y)$ where $g(y) = E[X \mid Y = y]$.

Tower property check: $E[E[X \mid Y]] = \sum_y E[X \mid Y = y] \cdot P(Y = y) = E[X]$.

## 5.2 Filtrations and Adapted Processes

**Definition 5.2** (Filtration)**.** A *filtration* on $(\Omega, \mathcal{F}, P)$ is an increasing sequence of sub-$\sigma$-algebras

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}.$$

Intuitively, $\mathcal{F}_n$ represents the information available up to time $n$.

**Definition 5.3** (Adapted Process)**.** A stochastic process $\{X_n\}_{n \geq 0}$ is *adapted* to filtration $\{\mathcal{F}_n\}$ if $X_n$ is $\mathcal{F}_n$-measurable for all $n \geq 0$. That is, the value of $X_n$ is determined by the information in $\mathcal{F}_n$.

**Definition 5.4** (Natural Filtration)**.** The *natural filtration* generated by process $\{X_n\}$ is

$$\mathcal{F}_n^X = \sigma(X_0, X_1, \ldots, X_n),$$

the smallest $\sigma$-algebra making $X_0, \ldots, X_n$ all measurable. By definition, $\{X_n\}$ is always adapted to its natural filtration.

## 5.3 Martingales: Definition and Examples

**Definition 5.5** (Martingale)**.** A stochastic process $\{M_n\}_{n\geq 0}$ adapted to $\{\mathcal{F}_n\}$ with $E[|M_n|] < \infty$ for all $n$ is called:

- A *martingale* if $E[M_{n+1} \mid \mathcal{F}_n] = M_n$ a.s. for all $n$.
- A *submartingale* if $E[M_{n+1} \mid \mathcal{F}_n] \geq M_n$ a.s. for all $n$.
- A *supermartingale* if $E[M_{n+1} \mid \mathcal{F}_n] \leq M_n$ a.s. for all $n$.

**Remark 5.2.** Intuitively: a martingale models a *fair game* (conditional expected future value equals present value). A submartingale models a *favorable* game (you expect to gain on average). A supermartingale models an *unfavorable* game.

Note: if $\{M_n\}$ is a martingale, then $E[M_n] = E[M_0]$ for all $n$ (by the tower property applied repeatedly).

**Example 5.2** (Simple Random Walk)**.** Let $X_1, X_2, \ldots$ be i.i.d. with $P(X_i = +1) = P(X_i = -1) = 1/2$. Define $S_n = X_1 + \cdots + X_n$ and $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$. Then:

$$E[S_{n+1} \mid \mathcal{F}_n] = E[S_n + X_{n+1} \mid \mathcal{F}_n] = S_n + E[X_{n+1}] = S_n + 0 = S_n.$$

So $\{S_n\}$ is a martingale with respect to $\{\mathcal{F}_n\}$.

**Example 5.3** (Product Martingale)**.** Let $X_1, X_2, \ldots$ be i.i.d. with $E[X_i] = 1$ (e.g., $P(X_i = 2) = P(X_i = 0) = 1/2$). Define $M_n = X_1 X_2 \cdots X_n$, $M_0 = 1$. Then:

$$E[M_{n+1} \mid \mathcal{F}_n] = M_n \cdot E[X_{n+1}] = M_n \cdot 1 = M_n.$$

So $\{M_n\}$ is a martingale.

**Example 5.4** (Doob Martingale)**.** Let $Z$ be an integrable random variable and $\{\mathcal{F}_n\}$ any filtration. Define

$$M_n = E[Z \mid \mathcal{F}_n].$$

Then $\{M_n\}$ is a martingale: by the tower property,

$$E[M_{n+1} \mid \mathcal{F}_n] = E[E[Z \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n] = E[Z \mid \mathcal{F}_n] = M_n.$$

This is called the *Doob martingale* or *Lévy's upward theorem* construction.

**Example 5.5** (Exponential Martingale)**.** For a simple random walk $S_n = \sum_{i=1}^n X_i$ with $X_i \in \{-1, +1\}$ equally likely, define for $\theta \in \mathbb{R}$:

$$M_n = \frac{e^{\theta S_n}}{(E[e^{\theta X_1}])^n} = \frac{e^{\theta S_n}}{(\cosh \theta)^n}.$$

One can verify $E[M_{n+1} \mid \mathcal{F}_n] = M_n$, so this is a martingale. Exponential martingales are key tools for obtaining concentration inequalities.

## 5.4 Stopping Times

**Definition 5.6** (Stopping Time)**.** A random variable $\tau : \Omega \to \{0, 1, 2, \ldots\} \cup \{\infty\}$ is a *stopping time* with respect to filtration $\{\mathcal{F}_n\}$ if

$$\{\tau \leq n\} \in \mathcal{F}_n \quad \text{for all } n \geq 0.$$

Equivalently, $\{\tau = n\} \in \mathcal{F}_n$ for all $n$. The event "we stop at or before time $n$" must be determined by information available at time $n$ alone.

**Remark 5.3.** Examples of stopping times: the first time a process hits a set $A$ (hitting time $\tau_A = \inf\{n : X_n \in A\}$), a fixed deterministic time $\tau = N$, and the minimum of two stopping times.

A *non-example*: "the last time $S_n = 0$ before time $N$" is NOT a stopping time, because determining it requires future information.

**Definition 5.7** (Stopped Process and $\sigma$-algebra $\mathcal{F}_\tau$)**.** For a stopping time $\tau$, the *stopped process* is $M_{\tau\wedge n} = M_{\min(\tau,n)}$.

The *$\sigma$-algebra at stopping time $\tau$* is

$$\mathcal{F}_\tau = \{A \in \mathcal{F} : A \cap \{\tau \leq n\} \in \mathcal{F}_n \text{ for all } n\}.$$

$\mathcal{F}_\tau$ represents the information accumulated up to the (random) stopping time.

**Lemma 5.1** (Stopped Martingale)**.** *If $\{M_n\}$ is a martingale and $\tau$ is a stopping time, then the stopped process $\{M_{\tau\wedge n}\}$ is also a martingale.*

*Proof.* By adaptedness and the martingale property:

$$E[M_{\tau\wedge(n+1)} \mid \mathcal{F}_n] = M_{\tau\wedge n} \cdot \mathbf{1}_{\tau\leq n} + E[M_{n+1} \mid \mathcal{F}_n] \cdot \mathbf{1}_{\tau>n} = M_{\tau\wedge n} \cdot \mathbf{1}_{\tau\leq n} + M_n \cdot \mathbf{1}_{\tau>n} = M_{\tau\wedge n}.$$

□

## 5.5 Optional Stopping Theorem

**Theorem 5.2** (Optional Stopping Theorem (Doob's OST))**.** *Let $\{M_n\}$ be a martingale and $\tau$ a stopping time. Then $E[M_\tau] = E[M_0]$ provided any one of the following conditions holds:*

*(i) $\tau$ is bounded: $\tau \leq N$ a.s. for some fixed $N < \infty$;*

*(ii) $\{M_n\}$ is uniformly integrable and $\tau < \infty$ a.s.;*

*(iii) $E[\tau] < \infty$ and $|M_{n+1} - M_n| \leq c$ a.s. (bounded increments).*

*Proof (bounded case).* Since $\tau \leq N$, we have $M_\tau = M_{\tau\wedge N}$. Since the stopped process is a martingale:

$$E[M_\tau] = E[M_{\tau\wedge N}] = E[M_0].$$

The last equality uses the martingale property of $M_{\tau\wedge n}$ iterated from $n = 0$ to $n = N$. □

**Remark 5.4.** The bounded case follows simply because we can write:

$$M_\tau = M_0 + \sum_{k=1}^{N} \mathbf{1}_{\tau \geq k}(M_k - M_{k-1}).$$

Taking expectations, each term $E[\mathbf{1}_{\tau \geq k}(M_k - M_{k-1})] = 0$ because $\{\tau \geq k\} = \{\tau \leq k-1\}^c \in \mathcal{F}_{k-1}$ and $E[M_k - M_{k-1} \mid \mathcal{F}_{k-1}] = 0$.

**Example 5.6** (Gambler's Ruin via OST)**.** Let $S_n = X_1 + \cdots + X_n$ be a simple random walk with $P(X_i = +1) = P(X_i = -1) = 1/2$. The gambler starts at position $s \in (0, a+b)$ and plays until hitting 0 (ruin) or $a+b$ (win), where the barrier positions are 0 and $a+b$.

More precisely: start at $S_0 = b$ (positive integer), stop at $\tau = \inf\{n : S_n = 0 \text{ or } S_n = a+b\}$. Let $p =$ probability of reaching $a+b$ before 0.

**Method 1:** $\{S_n\}$ is a martingale. Applying OST (condition (i) or (iii)):

$$E[S_\tau] = E[S_0] = b.$$

But $S_\tau \in \{0, a+b\}$, so:

$$E[S_\tau] = (a+b) \cdot p + 0 \cdot (1-p) = (a+b) \cdot p.$$

Therefore $p = b/(a+b)$.

**Method 2:** $\{S_n^2 - n\}$ is a martingale (since $E[S_{n+1}^2 \mid \mathcal{F}_n] = S_n^2 + 1$). Applying OST:

$$E[S_\tau^2 - \tau] = E[S_0^2 - 0] = b^2.$$

Since $E[S_\tau^2] = (a+b)^2 \cdot p = (a+b)^2 \cdot \frac{b}{a+b} = b(a+b)$, we get $E[\tau] = b(a+b) - b^2 = ab$.

## 5.6 Martingale Convergence

**Theorem 5.3** (Martingale Convergence Theorem)**.** *Let $\{M_n\}$ be a martingale (or submartingale) with $\sup_n E[|M_n|] < \infty$. Then there exists a random variable $M_\infty$ with $E[|M_\infty|] < \infty$ such that*

$$M_n \xrightarrow{a.s.} M_\infty \quad \text{as } n \to \infty.$$

**Remark 5.5.** Note that a.s. convergence does NOT necessarily imply $L^1$ convergence. For $L^1$ convergence (i.e., $E[|M_n - M_\infty|] \to 0$), we need the additional condition of *uniform integrability.*

**Definition 5.8** (Uniform Integrability)**.** A family $\{X_\alpha\}$ of random variables is *uniformly integrable (UI)* if

$$\lim_{K \to \infty} \sup_\alpha E\big[|X_\alpha| \mathbf{1}_{|X_\alpha| > K}\big] = 0.$$

Equivalently: the tails of all $X_\alpha$ are uniformly small in expectation.

**Theorem 5.4** ($L^1$ Martingale Convergence)**.** *A martingale $\{M_n\}$ converges in $L^1$ (and hence a.s.) if and only if it is uniformly integrable. In this case, $M_\infty = \lim_n M_n$ exists a.s. and $E[M_n \mid \mathcal{F}_m] = M_m$ for all $m \leq n$ extends to $E[M_\infty \mid \mathcal{F}_n] = M_n$.*

**Example 5.7** (Galton–Watson Branching Process and Martingale)**. Setup.** A population evolves in generations. Each individual in generation $n$ independently produces a random number of offspring according to a distribution with pgf $G(s) = \mathbb{E}[s^L]$ where $L$ counts offspring. Let $Z_0 = 1$, $Z_n =$ size of generation $n$, and $\mu = \mathbb{E}[L] = G'(1)$.

**The martingale.** Define $M_n = Z_n/\mu^n$ (normalized population size). Then $\{M_n\}$ is a non-negative martingale with respect to $\mathcal{F}_n = \sigma(Z_0, Z_1, \ldots, Z_n)$:

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = \frac{1}{\mu^{n+1}}\mathbb{E}[Z_{n+1} \mid Z_n] = \frac{1}{\mu^{n+1}} \cdot \mu Z_n = \frac{Z_n}{\mu^n} = M_n.$$

Here we used $\mathbb{E}[Z_{n+1} \mid Z_n = k] = k\mu$ (each of $k$ individuals produces mean $\mu$ offspring independently).

**Convergence.** Since $M_n \geq 0$ and $\mathbb{E}[M_n] = \mathbb{E}[M_0] = 1 < \infty$, by the Martingale Convergence Theorem, $M_n \to M_\infty$ a.s.

**Extinction probability.** Let $q = \mathbb{P}(Z_n = 0 \text{ for some } n) = \mathbb{P}(\text{extinction})$. Then $q$ is the smallest non-negative solution to $q = G(q)$:

- If $\mu = \mathbb{E}[L] \leq 1$ (subcritical or critical): $q = 1$ (certain extinction).
- If $\mu > 1$ (supercritical): $q < 1$ (survival with positive probability $1 - q$).

On the extinction event, $M_\infty = 0$. On survival ($\mu > 1$), $M_\infty > 0$ with positive probability, capturing the limiting normalized population size.

**Second moment and $L^2$ convergence.** If $\sigma^2 = \mathrm{Var}(L) < \infty$, then $\mathbb{E}[M_n^2]$ stays bounded (compute via $\mathbb{E}[Z_n^2]$ recursion), so $M_n \to M_\infty$ in $L^2$ as well. In particular, $\mathbb{E}[M_\infty] = 1 \cdot \mathbb{P}(\text{survival}) + 0 \cdot \mathbb{P}(\text{extinction})$, which with $L^2$ convergence gives $\mathbb{E}[M_\infty] = 1 - q$ only when $\mu > 1$ and $\sigma^2 < \infty$.

## 5.7 Azuma–Hoeffding Inequality

**Definition 5.9** (Martingale Difference Sequence)**.** Given a martingale $\{M_n\}$ with respect to $\{\mathcal{F}_n\}$, the *martingale differences* are $D_k = M_k - M_{k-1}$ for $k \geq 1$. Note that $E[D_k \mid \mathcal{F}_{k-1}] = 0$ and $M_n = M_0 + \sum_{k=1}^n D_k$.

**Definition 5.10** (Bounded Differences Condition)**.** A martingale $\{M_n\}$ satisfies the *bounded differences condition* with constants $c_1, c_2, \ldots > 0$ if

$$|D_k| = |M_k - M_{k-1}| \leq c_k \quad \text{a.s. for all } k \geq 1.$$

**Theorem 5.5** (Azuma–Hoeffding Inequality)**.** *Let $\{M_n\}_{n=0}^N$ be a martingale satisfying the bounded differences condition with constants $c_1, \ldots, c_N$. Then for any $t > 0$:*

$$P(M_N - M_0 \geq t) \leq \exp\left(-\frac{t^2}{2\sum_{k=1}^N c_k^2}\right),$$

*and similarly:*

$$P(|M_N - M_0| \geq t) \leq 2\exp\left(-\frac{t^2}{2\sum_{k=1}^N c_k^2}\right).$$

*Proof Sketch.* The proof uses the Hoeffding lemma: if $E[D] = 0$ and $|D| \le c$ a.s., then $E[e^{\lambda D}] \le e^{\lambda^2 c^2/2}$.

Define $L_n = e^{\lambda M_n} / \prod_{k=1}^n E[e^{\lambda D_k} \mid \mathcal{F}_{k-1}]$. Using the Hoeffding lemma and Markov's inequality on the exponential martingale $\{e^{\lambda M_n}\}$, we get:

$$P(M_N - M_0 \ge t) = P\big(e^{\lambda(M_N - M_0)} \ge e^{\lambda t}\big) \le e^{-\lambda t} \cdot E[e^{\lambda(M_N - M_0)}] \le e^{-\lambda t + \lambda^2 \sum c_k^2/2}.$$

Optimizing over $\lambda > 0$ (set $\lambda = t/\sum c_k^2$) gives the result. □

**Corollary 5.1** (McDiarmid's Inequality)**.** *Let $f(x_1, \ldots, x_n)$ be a function of $n$ independent random variables satisfying the bounded differences condition:*

$$\sup_{x_1,\ldots,x_n,\, x_i'} |f(\ldots, x_i, \ldots) - f(\ldots, x_i', \ldots)| \le c_i.$$

*Then with $Z = f(X_1, \ldots, X_n)$ and $\mu = E[Z]$:*

$$P(|Z - \mu| \ge t) \le 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

*This follows from Azuma–Hoeffding applied to the Doob martingale $M_k = E[Z \mid X_1, \ldots, X_k]$.*

**Example 5.8** (Azuma–Hoeffding Application)**.** Consider a simple random walk $S_n = X_1 + \cdots + X_n$ with $X_i \in \{-1, +1\}$. The differences are $|S_n - S_{n-1}| = 1$, so $c_k = 1$ for all $k$. By Azuma–Hoeffding:

$$P(S_n \ge t) \le e^{-t^2/(2n)}.$$

Equivalently, $P(S_n \ge t\sqrt{n}) \le e^{-t^2/2}$, consistent with the CLT approximation $S_n/\sqrt{n} \approx N(0, 1)$.

**Example 5.9** (Hamming Distance via Azuma–Hoeffding (PS13 type))**.** Let $X_1, \ldots, X_n \in \{0, 1\}$ be i.i.d. Bernoulli($p$) bits (the transmitted codeword), and let $f(X_1, \ldots, X_n)$ be the Hamming distance from the received word to the nearest codeword in some fixed code $\mathcal{C}$:

$$f(x_1, \ldots, x_n) = \min_{c \in \mathcal{C}} d_H(x, c), \quad d_H(x, c) = \sum_{i=1}^n \mathbf{1}[x_i \ne c_i].$$

Changing any single bit $x_i \to x_i'$ changes the Hamming distance by at most 1, so $f$ satisfies the bounded differences condition with $c_i = 1$ for all $i$. By McDiarmid's inequality:

$$P\big(|f(X_1, \ldots, X_n) - \mathbb{E}[f]| \ge t\big) \le 2e^{-2t^2/n}.$$

This says the minimum distance decoding error is concentrated around its mean: with high probability, $f$ is within $O(\sqrt{n \log(1/\delta)})$ of its expectation.

**Example 5.10** (Graph Chromatic Number / Shortest Path via Azuma–Hoeffding (PS13 type))**. Chromatic number:** Let $G(n, p)$ be the Erdős–Rényi random graph on $n$ vertices. Let $\chi(G)$ be the chromatic number. Reveal the edges one by one; revealing edge $\{i, j\}$ changes $\chi$ by at most 1. There are $\binom{n}{2}$ edges, each bounded by $c_k = 1$. By Azuma–Hoeffding:

$$P\big(|\chi(G) - \mathbb{E}[\chi(G)]| \ge t\big) \le 2 \exp\left(-\frac{2t^2}{\binom{n}{2}}\right) = 2e^{-4t^2/n(n-1)}.$$

**Shortest path:** Let $d(s, t)$ be the shortest path length between two fixed vertices $s, t$ in $G(n, p)$. Changing one edge changes $d(s, t)$ by at most 1, so:

$$P\big(|d(s, t) - \mathbb{E}[d(s, t)]| \ge t\big) \le 2e^{-4t^2/n(n-1)}.$$

These examples illustrate how Azuma–Hoeffding via the Doob martingale $M_k = \mathbb{E}[f \mid \text{first } k \text{ edges}]$ gives concentration for any Lipschitz function of random inputs.

## 5.8 Wald's Identity

**Theorem 5.6** (Wald's Identity)**.** *Let $X_1, X_2, \ldots$ be i.i.d. with $E[X_1] = \mu < \infty$, and let $\tau$ be a stopping time with respect to the natural filtration $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$, with $E[\tau] < \infty$. Define $S_n = X_1 + \cdots + X_n$. Then*

$$E[S_\tau] = \mu \cdot E[\tau].$$

***Conditions required for Wald's identity to hold:***

*(i)* $X_1, X_2, \ldots$ *are* ***i.i.d.*** *(identically distributed and mutually independent).*

*(ii)* $E[|X_1|] < \infty$ *(finite mean $\mu$).*

*(iii)* $\tau$ *is a* ***stopping time*** *— i.e., $\{\tau = n\} \in \mathcal{F}_n = \sigma(X_1, \ldots, X_n)$ for all $n$. Crucially, $\tau$ may depend on $X_1, X_2, \ldots$ but only through their history, not future values.*

*(iv)* $E[\tau] < \infty$.

*Proof.* Write $S_\tau = \sum_{k=1}^{\infty} X_k \mathbf{1}_{k \le \tau}$. Then:

$$E[S_\tau] = \sum_{k=1}^{\infty} E[X_k \mathbf{1}_{k \le \tau}].$$

Since $\{\tau \ge k\} = \{\tau \le k-1\}^c \in \mathcal{F}_{k-1}$ and $X_k$ is independent of $\mathcal{F}_{k-1}$:

$$E[X_k \mathbf{1}_{k \le \tau}] = E[X_k] \cdot P(\tau \ge k) = \mu \cdot P(\tau \ge k).$$

Summing: $E[S_\tau] = \mu \sum_{k=1}^{\infty} P(\tau \ge k) = \mu \cdot E[\tau]$. □

**Remark 5.6.** Alternatively: $M_n = S_n - n\mu$ is a martingale. By the Optional Stopping Theorem (under $E[\tau] < \infty$ and bounded increments),

$$E[M_\tau] = E[S_\tau - \tau\mu] = E[M_0] = 0 \implies E[S_\tau] = \mu E[\tau].$$

**Example 5.11** (Counterexample: Wald fails when $\tau$ is not a stopping time)**.** Let $X_1, X_2, \ldots$ be i.i.d. with $P(X_i = +1) = P(X_i = -1) = 1/2$, so $\mu = 0$.

**Failure case 1: $\tau$ depends on future values.** Define $\tau = \arg\max_{1 \le k \le n} S_k$ (the time index at which the maximum is achieved among the first $n$ steps). This is NOT a stopping time since determining whether $k$ is the maximizer requires knowledge of $X_{k+1}, \ldots, X_n$ (future information). Wald's formula gives $E[S_\tau] = \mu \cdot E[\tau] = 0$, but in fact $E[S_\tau] = E[\max_{k \le n} S_k] > 0$ for $n \ge 1$.

**Failure case 2: $E[\tau] = \infty$.** Let $X_i$ be i.i.d. with $P(X_i = 2) = P(X_i = -1) = 1/2$, so $\mu = 1/2$. Let $\tau = \min\{n : S_n \le 0\}$. One can show $E[\tau] = \infty$ in this case (because positive drift makes return to 0 very slow). Then $E[S_\tau] = E[S_\tau \mid S_\tau \le 0] \le 0$, but $\mu \cdot E[\tau] = \infty$, so Wald's identity fails.

**Failure case 3 (PS13 Q2 type): $\tau$ not independent of $X$'s through stopping time structure.** Suppose we define $\tau = n$ if $X_n = 1$ and $\tau = 1$ otherwise (i.e., $\tau$ is chosen to land on a favorable observation). This is not a stopping time (the choice of $\tau$ depends on $X_\tau$, which is future information at time $\tau - 1$). Computing directly: $E[S_\tau] \ne \mu \cdot E[\tau]$ in general.

**Key takeaway:** Wald's identity requires $\tau$ to be a genuine stopping time — determined by the past — and $E[\tau] < \infty$. If either condition fails, the identity can break down.

**Example 5.12** (Expected Stopping Time for Random Walk)**.** For the simple random walk (with $\mu = 0$) starting at $S_0 = b$ and stopping at $\{0, a + b\}$:

From the gambler's ruin calculation, $E[\tau] = ab$ (derived using $M_n = S_n^2 - n$ and OST).

As another application: for i.i.d. $X_i \in \{-1, +1\}$ with $P(X_i = +1) = p \neq 1/2$, define $\mu = 2p - 1 \neq 0$. Then $S_n - n\mu$ is a martingale. By Wald: if we stop at $\tau = \inf\{n : S_n \notin (0, N)\}$, then $E[S_\tau] = b + \mu \cdot E[\tau]$, giving $E[\tau]$ in terms of the boundary hitting probabilities.

## 5.9 Applications to Queueing

**Remark 5.7** (Martingales in Queueing Theory)**.** Martingale methods provide powerful tools for analyzing random processes in queueing:

- **Stability:** If $Q_n$ is the queue length process, $\{Q_n - \rho n\}$ for appropriate $\rho$ can be a supermartingale, establishing stability conditions via convergence theorems.
- **Waiting time bounds:** Azuma–Hoeffding applied to service time sums yields concentration bounds on waiting times.
- **Fluid models:** Doob martingales arise naturally in the analysis of fluid-flow approximations to queues.

**Example 5.13** (Martingale Analysis of M/G/1 Queue)**.** In an M/G/1 queue with Poisson arrivals (rate $\lambda$), service times $B_i$ (i.i.d., mean $1/\mu$, second moment $E[B^2]$), let $Q_n$ be the queue length just after the $n$-th departure.

The sequence $\{Q_n\}$ is a Markov chain. Define $\rho = \lambda/\mu < 1$ (stability condition). For the stationary analysis, a key martingale identity is the Pollaczek–Khinchine (P-K) formula:

$$E[Q] = \rho + \frac{\rho^2 + \lambda^2 E[B^2]}{2(1 - \rho)}.$$

The mean waiting time in system is $W = E[Q]/\lambda + 1/\mu$ (Little's Law).

## 5.10 Summary of Key Results

**Remark 5.8** (Course Summary)**.** The course IE622 covered the following major topics:

(i) **Renewal Theory:** Renewal processes, renewal reward theorem, key renewal theorem, alternating renewal processes.

(ii) **Markov Chains (DTMC):** Chapman–Kolmogorov equations, classification of states (transient/recurrent/positive recurrent), stationary distributions, ergodic theorem, birth-death chains.

(iii) **Markov Chains (CTMC):** Generator matrices, Kolmogorov forward/backward equations, stationary distributions, PASTA property, birth-death processes.

(iv) **Queueing Theory:** M/M/1, M/M/c, M/M/$\infty$ queues, Little's Law, Burke's theorem, tandem queues, Jackson networks with product-form solutions.

(v) **Martingale Theory:** Conditional expectation, filtrations, martingales/submartingales/superman stopping times, Optional Stopping Theorem, martingale convergence, Azuma–Hoeffding inequality, Wald's identity.

The unifying theme is the analysis of *stochastic processes* — random systems evolving over time — using tools from probability theory to derive exact or approximate performance measures.